A Short Path Joining Two Zeros Inside a Polynomial Lemniscate

1. Introduction

Let

$$f(z) = \prod_{j=1}^{n} (z - z_j), \qquad |z_j| < 1 \quad (1 \le j \le n),$$

where the zeros are listed with multiplicity. We consider the filled unit lemniscate

$$\Lambda_f := \{z \in \mathbb{C} : |f(z)| < 1\}.$$

The question addressed here is the following.

Must there always exist a rectifiable curve of Euclidean length less than 2, contained in Λ_f, whose endpoints are two roots of f?

Our main result answers this in the affirmative.

Theorem 1. *Let*

$$f(z) = \prod_{j=1}^{n} (z - z_j), \qquad |z_j| < 1 \quad (1 \le j \le n),$$

where the zeros are listed with multiplicity. Then there exist indices $i, j \in \{1, \dots, n\}$ and a rectifiable curve $\gamma \subset \Lambda_f$ joining z_i to z_j such that

$$\operatorname{len}(\gamma) < 2.$$

In particular, if f is squarefree and $n \ge 2$, then the endpoints may be chosen to be distinct zeros of f.

Remark 2. If f has a multiple zero, the conclusion is immediate: two coincident roots may be joined by the constant curve. Thus the only nontrivial case is when f is squarefree and $n \ge 2$. In the proof below we therefore reduce immediately to that case.

The main geometric input we use is the following component lemma.

Component lemma. Some connected component of Λ_f contains at least two zeros of f.

This is known due to work of Erdős, Herzog, and Piranian; see [2] and the later references listed below. In addition, we use standard results from planar potential theory, planar topology, and Morse theory: Carleman's inequality, Pólya's inequality, the Jordan–Schoenflies theorem, and the local Morse normal form for a nondegenerate saddle. The proof of Theorem 1 has three parts.

(1) For any component $U \subset \Lambda_f$ containing m zeros of f, we prove the sharp integral estimate

$$\int_U \frac{|f'(z)|}{|f(z)|}\, dA(z) \le 2\pi m.$$

(2) Above a fixed base level we construct, under a genericity hypothesis on the critical points, a finite tree spanning the relevant roots, with total length controlled by the tail integral of the level lengths.

(3) A small explicit perturbation of the form $f \mapsto f + \lambda z + \beta$ makes the critical-point configuration generic on a fixed collar, while preserving the relevant component and a fixed positive slack below the threshold 2.

2. An analytic estimate on a component

Let U be a connected component of Λ_f, and suppose that U contains exactly $m \geq 1$ zeros of f, counted with multiplicity. Set

$$u := -\log|f| \quad \text{on } U.$$

Then $u > 0$ on U, $u = 0$ on ∂U, and $u(z) \to +\infty$ at the zeros of f in U. For $t \geq 0$ define

$$K_t := \{z \in U : u(z) \geq t\}, \qquad A(t) := \mathrm{Area}(K_t).$$

For regular t write

$$P(t) := \mathcal{H}^1(\{z \in U : u(z) = t\}).$$

We also set

$$I(U) := \int_U \frac{|f'(z)|}{|f(z)|}\, dA(z) = \int_U |\nabla u|\, dA.$$

Lemma 3. *Let p be a nonconstant polynomial and let $r > 0$. Then the level set*

$$\{z \in \mathbb{C} : |p(z)| = r\}$$

has planar Lebesgue measure zero. Consequently, if E is a connected component of

$$\{z \in \mathbb{C} : |p(z)| < r\},$$

then $\partial E \subset \{|p| = r\}$ and

$$\mathrm{Area}(E) = \mathrm{Area}(\overline{E}).$$

Proof. Set

$$S := \{z \in \mathbb{C} : |p(z)| = r,\ p'(z) \neq 0\}.$$

The function $F(z) := |p(z)|^2 - r^2$ is real-analytic on $\mathbb{C}$, and on S its gradient is nonzero because, after identifying $\mathbb{C}$ with $\mathbb{R}^2$,

$$\nabla F(z) = 2\,\overline{p(z)}\,p'(z).$$

Therefore the implicit function theorem shows that S is a one-dimensional real-analytic submanifold of $\mathbb{C}$, hence it has planar measure zero. The complement

$$\{|p| = r\} \setminus S$$

is contained in the finite set $\{p' = 0\}$, because $r > 0$ excludes the zeros of p. Thus $\{|p| = r\}$ has planar measure zero.

If E is a connected component of $\{|p| < r\}$, then $\partial E \subset \{|p| = r\}$. Hence

$$\mathrm{Area}(\overline{E} \setminus E) = 0,$$

so $\mathrm{Area}(E) = \mathrm{Area}(\overline{E})$. $\square$

Auxiliary lemma (flux across a regular superlevel component). Let $s > 0$ be a regular value of $u = -\log|f|$, and let G be a connected component of

$$\{z \in U : u(z) > s\}.$$

Suppose that the zeros of f in G are $a_1, \dots, a_r$, with multiplicities $m_1, \dots, m_r$, and set

$$M := m_1 + \cdots + m_r.$$

Then

$$\int_{\partial G} |\nabla u| \, ds = 2\pi M.$$

Proof. Because s is a regular value, $\partial G \subset \{u = s\}$ is a smooth compact 1-manifold. For $\rho > 0$ small enough, let

$$G_\rho := G \setminus \bigcup_{j=1}^{r} \overline{D(a_j, \rho)}.$$

Since u is harmonic on G_ρ, Green's identity gives

$$0 = \int_{G_\rho} \Delta u \, dA = \int_{\partial G_\rho} \partial_\nu u \, ds = \int_{\partial G} \partial_\nu u \, ds + \sum_{j=1}^{r} \int_{\partial D(a_j,\rho)} \partial_\nu u \, ds,$$

where ν denotes the outward unit normal of G_ρ.

On ∂G, the function u decreases in the outward direction, so

$$\partial_\nu u = -|\nabla u|.$$

Near a_j one has

$$u(z) = -m_j \log|z - a_j| + h_j(z),$$

where h_j is harmonic. Therefore, on $\partial D(a_j, \rho)$,

$$\partial_\nu u = \frac{m_j}{\rho} + O(1),$$

because the outward normal of G_ρ on the inner boundary points toward the center a_j. Hence

$$\int_{\partial D(a_j,\rho)} \partial_\nu u \, ds \to 2\pi m_j \qquad (\rho \downarrow 0).$$

Letting $\rho \downarrow 0$ yields

$$-\int_{\partial G} |\nabla u| \, ds + 2\pi \sum_{j=1}^{r} m_j = 0,$$

which is exactly the stated identity. □

Proposition 4. *With the notation above,*

$$I(U) \le 2\pi m.$$

Proof. Let $a_1, \dots, a_r$ be the distinct zeros of f in U, with multiplicities $m_1, \dots, m_r$, so that

$$m_1 + \cdots + m_r = m.$$

We first justify the coarea identities by truncation away from the singularities of $u = -\log|f|$.

Fix regular values $0 < \varepsilon < t$, and set

$$U_\varepsilon := \{z \in U : u(z) > \varepsilon\}, \qquad \Omega_{\varepsilon,t} := U_\varepsilon \setminus K_t = \{\varepsilon < u < t\}.$$

Since every zero of f in U lies in K_t, the function u is harmonic on a neighborhood of $\Omega_{\varepsilon,t}$. Consider

$$v_{\varepsilon,t}(z) := \min\left\{\frac{u(z)-\varepsilon}{t-\varepsilon},\, 1\right\} \qquad (z \in U_\varepsilon).$$

Then $v_{\varepsilon,t} = 0$ on ∂U_ε, $v_{\varepsilon,t} = 1$ on K_t, and

$$\nabla v_{\varepsilon,t} = \frac{\nabla u}{t-\varepsilon} \qquad \text{on } \Omega_{\varepsilon,t}.$$

Thus $v_{\varepsilon,t}$ is admissible for the condenser (K_t, U_ε).

We now justify directly that $v_{\varepsilon,t}$ minimizes the Dirichlet energy. Let w be any admissible competitor for (K_t, U_ε). Replacing w by $\min\{1, \max\{w, 0\}\}$ if necessary, we may assume $0 \le w \le 1$. On $\Omega_{\varepsilon,t}$ set

$$\eta := w - v_{\varepsilon,t}.$$

Since both w and $v_{\varepsilon,t}$ equal 0 on ∂U_ε and equal 1 on $\partial K_t = \{u = t\}$, the function η has zero trace on $\partial\Omega_{\varepsilon,t}$. Because u is harmonic on $\Omega_{\varepsilon,t}$, so is $v_{\varepsilon,t}$. Hence

$$\int_{\Omega_{\varepsilon,t}} |\nabla w|^2\, dA = \int_{\Omega_{\varepsilon,t}} |\nabla v_{\varepsilon,t}|^2\, dA + \int_{\Omega_{\varepsilon,t}} |\nabla \eta|^2\, dA + 2\int_{\Omega_{\varepsilon,t}} \nabla v_{\varepsilon,t} \cdot \nabla \eta\, dA.$$

The last term vanishes by integration by parts:

$$\int_{\Omega_{\varepsilon,t}} \nabla v_{\varepsilon,t} \cdot \nabla \eta\, dA = -\int_{\Omega_{\varepsilon,t}} \eta\, \Delta v_{\varepsilon,t}\, dA + \int_{\partial\Omega_{\varepsilon,t}} \eta\, \partial_\nu v_{\varepsilon,t}\, ds = 0,$$

because $\Delta v_{\varepsilon,t} = 0$ on $\Omega_{\varepsilon,t}$ and $\eta|_{\partial\Omega_{\varepsilon,t}} = 0$. Therefore

$$\int_{\Omega_{\varepsilon,t}} |\nabla w|^2\, dA \ge \int_{\Omega_{\varepsilon,t}} |\nabla v_{\varepsilon,t}|^2\, dA.$$

Since the total Dirichlet energy of w on U_ε is at least its energy on $\Omega_{\varepsilon,t}$, it follows that $v_{\varepsilon,t}$ is the capacitary potential of (K_t, U_ε). With the normalization

$$\operatorname{cap}(E, B) = \frac{1}{4\pi} \inf \int |\nabla v|^2\, dA,$$

we obtain

$$\operatorname{cap}(K_t, U_\varepsilon) = \frac{1}{4\pi} \int_{\Omega_{\varepsilon,t}} |\nabla v_{\varepsilon,t}|^2\, dA = \frac{1}{4\pi(t-\varepsilon)^2} \int_{\varepsilon<u<t} |\nabla u|^2\, dA.$$

Applying the coarea formula on the smooth region $\Omega_{\varepsilon,t}$, we obtain

$$\int_{\varepsilon<u<t} |\nabla u|^2\, dA = \int_\varepsilon^t \left(\int_{u=s} |\nabla u|\, d\sigma\right) ds.$$

For each regular $s \in (\varepsilon, t)$, the level set $\{u = s\}$ is the disjoint union of the boundaries of the connected components of $\{u > s\}$. Applying the auxiliary flux lemma above to each such component and summing over all components shows that

$$\int_{u=s} |\nabla u|\, d\sigma = 2\pi m,$$

because the total multiplicity of the zeros of f in U is m. Hence

$$\int_{\varepsilon<u<t} |\nabla u|^2\, dA = \int_\varepsilon^t 2\pi m\, ds = 2\pi m(t-\varepsilon).$$

Therefore

$$\operatorname{cap}(K_t, U_\varepsilon) = \frac{m}{2(t-\varepsilon)}.$$

As $\varepsilon \downarrow 0$, the domains U_ε increase to U. We claim that

$$\operatorname{cap}_{\text{cond}}(K_t, U) = \lim_{\varepsilon \downarrow 0} \operatorname{cap}_{\text{cond}}(K_t, U_\varepsilon).$$

Indeed, for each $\varepsilon > 0$, every admissible function for (K_t, U_ε) is also admissible for (K_t, U), hence

$$\operatorname{cap}_{\text{cond}}(K_t, U) \le \operatorname{cap}_{\text{cond}}(K_t, U_\varepsilon).$$

Conversely, let $v \in C_c^\infty(U)$ satisfy $v \ge 1$ on K_t. Since $\operatorname{supp} v$ is compact in U, there exists $\varepsilon_0 > 0$ such that $\operatorname{supp} v \subset U_\varepsilon$ for all $0 < \varepsilon < \varepsilon_0$. Thus, for all such ε,

$$\operatorname{cap}_{\text{cond}}(K_t, U_\varepsilon) \le \frac{1}{4\pi} \int_{\mathbb{C}} |\nabla v|^2 \, dA.$$

Taking $\limsup$ as $\varepsilon \downarrow 0$ and then the infimum over all such v gives the reverse inequality. Therefore

$$\operatorname{cap}_{\text{cond}}(K_t, U) = \frac{m}{2t}. \tag{5}$$

For bounded open sets $G \subset \mathbb{C}$ we write

$$\operatorname{cap}_{\log}(G) := \operatorname{cap}_{\log}(\overline{G}).$$

We also record explicitly the normalization used here: for the round condenser

$$\Big(D(0,R), \overline{D(0,r)}\Big) \qquad (0 < r < R),$$

$$\operatorname{cap}_{\text{cond}}\Big(\overline{D(0,r)}, D(0,R)\Big) = \frac{1}{2\log(R/r)}.$$

Accordingly, Carleman's inequality in the present normalization reads

$$\frac{1}{\operatorname{cap}_{\text{cond}}(E,B)} \le \log \frac{\operatorname{Area}(B)}{\operatorname{Area}(E)} \qquad (E \Subset B),$$

and Pólya's inequality reads

$$\operatorname{Area}(K) \le \pi \operatorname{cap}_{\log}(K)^2 \qquad (K \subset \mathbb{C} \text{ compact}).$$

Applying Carleman's inequality to the condenser (U, K_t) and using (5), we obtain

$$\log \frac{\operatorname{Area}(U)}{A(t)} \ge \frac{2t}{m},$$

that is,

$$A(t) \le \operatorname{Area}(U) e^{-2t/m}. \tag{6}$$

It remains to bound $\operatorname{Area}(U)$. Since f is monic, the filled lemniscate

$$\{z \in \mathbb{C} : |f(z)| \le 1\}$$

has logarithmic capacity 1. Hence, by monotonicity of logarithmic capacity,

$$\operatorname{cap}_{\log}(\overline{U}) \le 1.$$

By Lemma 3, $\partial U \subset \{|f| = 1\}$ has planar measure zero, so

$$\operatorname{Area}(U) = \operatorname{Area}(\overline{U}).$$

Applying Pólya's inequality to $\overline{U}$ yields

$$\text{Area}(U) = \text{Area}(\overline{U}) \leq \pi \ \text{cap}_{\log}(\overline{U})^2 \leq \pi. \tag{7}$$

Combining (6) and (7), we get

$$A(t) \leq \pi e^{-2t/m}.$$

We now record explicitly the regularity of the distribution function

$$A(t) = \text{Area}(K_t).$$

Since u has only finitely many critical values and is smooth away from the zeros of f, the function A is absolutely continuous on every compact interval of $(0, \infty)$. Moreover, for almost every regular value $t > 0$,

$$-A'(t) = \int_{u=t} \frac{1}{|\nabla u|} \, ds.$$

Indeed, if $[a, b]$ contains no critical values of u, then the coarea formula applied on $\{a < u < b\}$ gives

$$A(a) - A(b) = \int_{\{a<u<b\}} 1 \, dA = \int_a^b \left(\int_{u=t} \frac{1}{|\nabla u|} \, ds \right) dt.$$

Since there are only finitely many critical values, this yields the asserted absolute continuity and derivative formula for almost every $t > 0$.

For almost every regular $t > 0$, Cauchy–Schwarz and the flux identity give

$$P(t)^2 = \left(\int_{u=t} 1 \, ds \right)^2 \leq \left(\int_{u=t} |\nabla u| \, ds \right) \left(\int_{u=t} \frac{1}{|\nabla u|} \, ds \right) = 2\pi m \, (-A'(t)).$$

Hence

$$P(t) \leq \sqrt{2\pi m \, (-A'(t))}. \tag{8}$$

Using the elementary inequality

$$2\sqrt{xy} \leq \frac{x}{\lambda} + \lambda y$$

with $x = 2\pi m$, $y = -A'(t)$, and $\lambda = m e^{t/m}$, we obtain

$$P(t) \leq \pi e^{-t/m} + \frac{m}{2} e^{t/m} (-A'(t)).$$

Integrating and using (1),

$$I(U) \leq \pi \int_0^\infty e^{-t/m} \, dt + \frac{m}{2} \int_0^\infty e^{t/m} (-A'(t)) \, dt.$$

The first integral equals πm. For the second, the bound $A(t) \leq \pi e^{-2t/m}$ implies $e^{t/m} A(t) \leq \pi e^{-t/m} \to 0$ as $t \to \infty$, so integration by parts gives

$$\int_0^\infty e^{t/m} (-A'(t)) \, dt = A(0) + \frac{1}{m} \int_0^\infty e^{t/m} A(t) \, dt.$$

Using (7) and (6),

$$A(0) \leq \pi, \qquad \frac{1}{m} \int_0^\infty e^{t/m} A(t) \, dt \leq \frac{\pi}{m} \int_0^\infty e^{-t/m} \, dt = \pi.$$

Therefore

$$\int_0^\infty e^{t/m}(-A'(t))\,dt \le 2\pi,$$

and hence

$$I(U) \le \pi m + \frac{m}{2}(2\pi) = 2\pi m.$$

□

Corollary 5. *Let h be a monic polynomial, let $c \ge 0$, and let $v := -\log|h|$. Let W be a connected component of $\{v > c\}$. If W contains exactly M zeros of h, counted with multiplicity, then*

$$\int_W |\nabla v|\,dA \le 2\pi M.$$

Equivalently,

$$\frac{1}{2\pi}\int_c^\infty P_W(t)\,dt \le M,$$

where $P_W(t) = \mathcal{H}^1(\{z \in W : v(z) = t\})$ for regular $t > c$.

Proof. Set

$$\widetilde{h} := e^c h, \qquad \widetilde{v} := -\log|\widetilde{h}| = v - c.$$

Then W is a connected component of

$$\{\widetilde{v} > 0\} = \{|\widetilde{h}| < 1\},$$

and W contains exactly M zeros of $\widetilde{h}$.

The proof of Proposition 4 applies verbatim to $\widetilde{h}$ on W. The only place where monicity was used there was the bound $\mathrm{Area}(W) \le \pi$, obtained via Pólya's inequality. Here the leading coefficient of $\widetilde{h}$ is e^c, so the filled lemniscate $\{|\widetilde{h}| \le 1\}$ has logarithmic capacity $e^{-c/\deg h} \le 1$. Since $W \subset \{|\widetilde{h}| < 1\}$, monotonicity gives

$$\mathrm{cap}_{\log}(W) = \mathrm{cap}_{\log}(\overline{W}) \le 1.$$

By Lemma 3, the boundary $\partial W \subset \{|\widetilde{h}| = 1\}$ has planar measure zero, so

$$\mathrm{Area}(W) = \mathrm{Area}(\overline{W}).$$

Applying Pólya's inequality to $\overline{W}$ yields

$$\mathrm{Area}(W) = \mathrm{Area}(\overline{W}) \le \pi\ \mathrm{cap}_{\log}(W)^2 \le \pi.$$

Therefore the same argument gives

$$\int_W |\nabla \widetilde{v}|\,dA \le 2\pi M.$$

Since $\nabla\widetilde{v} = \nabla v$, this is the desired estimate. □

3. The first critical value and the low superlevel sets

Throughout the rest of the paper, f is assumed squarefree on the component under consideration. Let $U \subset \Lambda_f$ be a connected component containing exactly $m \geq 2$ zeros of f, and write again

$$u = -\log|f| \quad \text{on } U.$$

Lemma 6. *The function u has at least one critical point in U. Consequently the set of positive critical values of u is nonempty, and it has a smallest element. Denote that smallest positive critical value by λ_* .*

Proof. Choose pairwise disjoint closed disks $E_1, \dots, E_m \Subset U$ centered at the zeros of f. Since the superlevel sets of u shrink to the zeros of f as the level tends to $+\infty$, there exists $T > 0$ such that

$$K_T := \{u \geq T\} \subset \bigcup_{j=1}^{m} \operatorname{int}(E_j).$$

In particular $u < T$ on $\bigcup_j \partial E_j$, and K_T has at least m connected components, one contained in each E_j.

Assume for contradiction that u has no critical point in $\{0 < u \leq T\}$. Consider the vector field

$$X := \frac{\nabla u}{|\nabla u|^2}$$

on $\{0 < u \leq T\}$; it satisfies $Xu = 1$. Let $z \in U \setminus K_T$, and write $s := u(z) \in (0, T)$. Set

$$M_s := U \cap \{e^{-T} \leq |f| \leq e^{-s/2}\}.$$

The set M_s is compact. It is bounded because $M_s \subset \Lambda_f$ and Λ_f is bounded. To see that it is closed in $\mathbb{C}$, let $z_k \in M_s$ with $z_k \to z$. Then continuity of f gives

$$e^{-T} \leq |f(z)| \leq e^{-s/2} < 1.$$

Hence $z \in \Lambda_f$. Since U is a connected component of the open set Λ_f, it is relatively closed in Λ_f; because z is also a limit point of U, we obtain $z \in U$. Thus $z \in M_s$. Also $M_s \subset \{s/2 \leq u \leq T\}$, so X is smooth on a neighborhood of M_s. Let $\varphi_\tau(z)$ be the maximal trajectory of X through z. Along this trajectory,

$$\frac{d}{d\tau} u(\varphi_\tau(z)) = Xu(\varphi_\tau(z)) = 1,$$

so

$$u(\varphi_\tau(z)) = s + \tau$$

whenever the trajectory is defined. In particular, for $0 \leq \tau \leq T - s$ one has

$$s/2 \leq s + \tau \leq T,$$

so the trajectory stays in the compact set M_s. Standard ODE continuation therefore shows that $\varphi_\tau(z)$ is defined on the whole interval $[0, T - s]$, and $\varphi_{T-s}(z) \in K_T$.

Define

$$R_T(z) = \begin{cases} z, & z \in K_T, \\ \varphi_{T-u(z)}(z), & z \in U \setminus K_T, \end{cases}$$

and

$$H : [0,1] \times U \to U, \qquad H(\theta, z) = \begin{cases} z, & z \in K_T, \\ \varphi_{\theta(T-u(z))}(z), & z \in U \setminus K_T. \end{cases}$$

If $z_0 \in U \setminus K_T$, then the trajectory segment

$$\{\varphi_\tau(z_0) : 0 \le \tau \le T - u(z_0)\}$$

is contained in the compact set $M_{u(z_0)}$, on which X is smooth. Standard continuous dependence for smooth ODEs therefore shows that the map

$$(\theta, z) \mapsto \varphi_{\theta(T-u(z))}(z)$$

is continuous near each point of $[0,1] \times (U \setminus K_T)$. At points with $u(z) = T$ the two formulas agree, so H is continuous on all of $[0,1] \times U$. Also

$$H(0,z) = z, \qquad H(1,z) = R_T(z), \qquad H(\theta, z) = z \text{ for every } z \in K_T.$$

Thus $R_T : U \to K_T$ is a well-defined strong deformation retraction. In particular K_T is connected, contradicting the previous paragraph. Therefore u has a critical point in $\{0 < u \le T\}$, and the set of positive critical values is nonempty.

Since $u = -\log|f|$ is smooth on $U \setminus Z(f)$, its finite critical points are exactly the points of $U \setminus Z(f)$ where $f'(z) = 0$. Because f is squarefree on U, we have

$$Z(f) \cap Z(f') = \varnothing.$$

Hence u has only finitely many finite critical points in U, and therefore only finitely many positive critical values. Thus the set of positive critical values has a smallest element, denoted λ_*. □

Set

$$\alpha := \lambda_*/4.$$

Then u has no critical values in $[0, 4\alpha)$.

Lemma 7. *For every $t \in [0, \lambda_*)$, the superlevel set $K_t := \{u \ge t\}$ is connected. In particular,*

$$K := \{u \ge \alpha/2\}, \qquad L := \{u \ge 3\alpha/2\}$$

are compact and connected.

Proof. Fix $0 < t < \lambda_*$. By definition of λ_*, the function u has no critical points in $\{0 < u \le t\}$. For $z \in U \setminus K_t$ write $s := u(z) \in (0,t)$, and set

$$M_s := U \cap \{e^{-t} \le |f| \le e^{-s/2}\}.$$

As in the proof of Lemma 6, M_s is compact and contained in $\{s/2 \le u \le t\}$, so the vector field

$$X = \frac{\nabla u}{|\nabla u|^2}$$

is smooth on a neighborhood of M_s. Along the maximal trajectory $\varphi_\tau(z)$ of X through z one has $u(\varphi_\tau(z)) = s + \tau$, hence the trajectory stays in M_s for $0 \le \tau \le t - s$ and therefore exists on that whole interval. Consequently

$$R_t(z) = \begin{cases} z, & z \in K_t, \\ \varphi_{t-u(z)}(z), & z \in U \setminus K_t, \end{cases}$$

is well defined. Define

$$H : [0,1] \times U \to U, \qquad H(\sigma, z) := \begin{cases} z, & z \in K_t, \\ \varphi_{\sigma(t-u(z))}(z), & z \in U \setminus K_t. \end{cases}$$

As in Lemma 6, the trajectory segments remain in the compact slab M_s, so standard continuous dependence for smooth ODEs implies that H is continuous. Thus H is a strong deformation retraction of U onto K_t. Since U is connected, K_t is connected. The case $t = 0$ is immediate.

For any $t > 0$,

$$K_t = U \cap \{|f| \le e^{-t}\},$$

because $|f| = 1$ on ∂U. The right-hand side is closed and bounded in $\mathbb{C}$, hence compact. Taking $t = \alpha/2$ and $t = 3\alpha/2$ gives the compactness of K and L. □

Define

$$q := \frac{1}{2\pi} \int_{\alpha}^{2\alpha} P(t)\, dt.$$

Since $P(t) > 0$ for $\alpha < t < 2\alpha$, we have $q > 0$. By Corollary 5,

$$\frac{1}{2\pi} \int_{2\alpha}^{\infty} P(t)\, dt \le m - q. \tag{9}$$

4. A generic spanning tree above the base level

Lemma 8 (Connected regular regions are strips)**.** *Let $0 < a < b$, and let $V \subset U$ be a connected open set such that:*

(i) *$a < u < b$ on V;*
(ii) *u has no critical points on V;*
(iii) *every component of*

$$\partial V \setminus \Big(\{u = a\} \cup \{u = b\}\Big)$$

is contained in an integral curve of

$$X := \frac{\nabla u}{|\nabla u|^2}.$$

For $t \in (a, b)$ set

$$\Sigma_t := V \cap \{u = t\}.$$

Then for every $s \in (a, b)$ the flow of X defines a diffeomorphism

$$\Phi_s : \Sigma_s \times (a, b) \to V, \qquad \Phi_s(x, t) = \varphi_{t-s}(x),$$

where φ_τ denotes the time-τ flow of X. In particular:

(a) *every level section Σ_t is connected;*
(b) *$\partial V \cap \{u = a\}$ is connected;*
(c) *$\partial V \cap \{u = b\}$ is connected.*

Thus V is a genuine strip with one lower cross-section and one upper cross-section.

Proof. Since $Xu = 1$, the function u increases at unit speed along each trajectory of X. Fix $s \in (a, b)$ and $x \in \Sigma_s$. Let $\gamma_x(\tau) = \varphi_\tau(x)$ be the maximal trajectory of X through x.

We first claim that $\gamma_x(\tau)$ is defined for all $\tau \in (a - s, b - s)$ and remains in V. Fix

$$0 < \eta < \min\{s - a, b - s\},$$

and let K_η be the closure in $\mathbb{C}$ of the set $V \cap \{a+\eta < u < b-\eta\}$. The set K_η is compact. Indeed, since $0 < a < b$ and $0 < \eta < \min\{s - a, b - s\}$, we have $a + \eta > 0$, so

$$V \cap \{a + \eta < u < b - \eta\} \subset \{e^{-(b-\eta)} \le |f| \le e^{-(a+\eta)}\},$$

and the set on the right is closed and bounded in $\mathbb{C}$ because Λ_f is bounded. Hence K_η, being the closure of a subset of a compact set, is compact. Every point of K_η lies in the interior of U: interior points of V are obvious, while points of $K_\eta \cap \partial V$ lie on the side boundary of V and satisfy $a+\eta \le u \le b-\eta$, so they cannot lie on $\partial U = \{u = 0\}$; hence they also lie in the interior of U. By hypothesis each side-boundary component is an integral curve of X, so $\nabla u \neq 0$ there as well. Therefore X is smooth on a neighborhood of K_η.

Along the trajectory one has

$$u(\gamma_x(\tau)) = s + \tau$$

whenever $\gamma_x(\tau)$ is defined. Thus, as long as

$$a + \eta \le u(\gamma_x(\tau)) \le b - \eta,$$

the trajectory stays in K_η. Standard ODE continuation then implies that the trajectory cannot cease to exist while $u(\gamma_x(\tau))$ remains in $[a + \eta, b - \eta]$.

The only remaining way in which the trajectory could fail to stay in V would be to hit the side boundary

$$\partial V \setminus \Big(\{u = a\} \cup \{u = b\}\Big)$$

at some first time $\tau_0 \in (a - s, b - s)$. Let $p = \gamma_x(\tau_0)$. By hypothesis, the side-boundary component through p is itself an integral curve of X. Since both that boundary curve and γ_x solve the same ODE and pass through p, uniqueness for the initial-value problem forces them to agree in a neighborhood of τ_0. But the boundary curve lies in ∂V, whereas $\gamma_x(\tau) \in V$ for $\tau < \tau_0$ close to τ_0, a contradiction. Hence $\gamma_x(\tau)$ never hits the side boundary.

Since $u(\gamma_x(\tau)) = s + \tau$, the preceding paragraph shows that $\gamma_x(\tau)$ exists and lies in V for every $\tau \in (a - s, b - s)$.

Therefore the formula

$$\Phi_s(x, t) = \varphi_{t-s}(x)$$

is well defined on $\Sigma_s \times (a, b)$. It is smooth. If $y \in V$, then the same compact-trapping and first-hit argument, with $u(y)$ in place of s, shows that the trajectory through y exists on the interval $[s - u(y), 0]$ and stays in V. Hence $\varphi_{s-u(y)}(y) \in \Sigma_s$, and the inverse map is

$$y \mapsto \Big(\varphi_{s-u(y)}(y), u(y)\Big),$$

which is smooth as well. So Φ_s is a diffeomorphism.

Since $V \cong \Sigma_s \times (a, b)$ and V is connected, the fiber Σ_s must be connected. This proves (a). For $\varepsilon > 0$ define

$$V_\varepsilon^- := V \cap \{a < u < a + \varepsilon\}, \qquad C_\varepsilon^- := \overline{V_\varepsilon^-},$$

where the closure is taken in $\mathbb{C}$. By the product representation, $V_\varepsilon^- = \Phi_s(\Sigma_s \times (a, a+\varepsilon))$ is connected, so C_ε^- is connected. Since V is bounded, each C_ε^- is compact. Moreover the family is nested:

$$C_{\varepsilon_2}^- \subset C_{\varepsilon_1}^- \qquad (0 < \varepsilon_2 < \varepsilon_1),$$

and

$$\bigcap_{\varepsilon>0} C_\varepsilon^- = \partial V \cap \{u = a\}.$$

Hence $\partial V \cap \{u = a\}$ is connected as an intersection of nested compact connected sets. The proof of (c) is identical, using

$$V_\varepsilon^+ := V \cap \{b - \varepsilon < u < b\}.$$

□

Lemma 9 (Regular lemniscate components are Jordan domains)**.** *Let c be a regular value of $u = -\log|f|$, and let W be a connected component of*

$$\{u > c\} = \{|f| < e^{-c}\}.$$

Then W is simply connected, and ∂W is a smooth Jordan curve.

Proof. Since c is a regular value of u, the set $\partial W \subset \{u = c\}$ is a smooth compact 1-manifold, hence a finite disjoint union of smooth Jordan curves.

Suppose that W had more than one boundary component. Choose an innermost boundary component $\Gamma \subset \partial W$ in the following sense: if H is the bounded Jordan domain enclosed by Γ, then H contains no other boundary component of any connected component of $\{u > c\}$. Such a choice is possible because the boundary consists of finitely many disjoint Jordan curves.

By the choice of Γ, the interior H contains no point of $\{u > c\}$. Otherwise some connected component of $\{u > c\}$ would lie in H, and one of its boundary components would be strictly inside Γ, contradicting innermostness. Hence

$$H \subset \{u \le c\} = \{|f| \ge e^{-c}\}, \qquad \partial H = \Gamma \subset \{u = c\}.$$

In particular f has no zeros on H, so

$$v := \log|f|$$

is harmonic on H and continuous on $\overline{H}$. On ∂H we have $v \equiv -c$. By the maximum principle, $v \equiv -c$ on H. Thus $|f|$ is constant on H, impossible for a nonconstant holomorphic function.

This contradiction shows that ∂W has only one connected component. Since it is a smooth compact 1-manifold, it is a single smooth Jordan curve. Therefore W is simply connected. □

Lemma 10 (Regular-slab deformation and connectivity)**.** *Let V be a connected component of $\{u > c\}$, and let $c < a < b$. Assume that u has no critical points on*

$$M := V \cap \{a \le u \le b\}.$$

Write

$$V_{\ge t} := V \cap \{u \ge t\}, \qquad V_{>t} := V \cap \{u > t\}.$$

Then:

(a) *the map*

$$R_{a,b} : V_{\geq a} \to V_{\geq b}, \qquad R_{a,b}(z) = \begin{cases} z, & u(z) \geq b, \\ \varphi_{b-u(z)}(z), & a \leq u(z) < b, \end{cases}$$

is a well-defined strong deformation retraction;

(b) *if $V_{\geq b}$ is connected, then $V_{>a}$ is connected. More precisely, every point of $V_{>a}$ can be joined to $V_{\geq b}$ by a trajectory segment of $X = \nabla u/|\nabla u|^2$ lying entirely in $V_{>a}$;*

(c) *if moreover a and b are regular values, then every connected component of $V \cap \{a < u < b\}$ is a regular strip in the sense of Lemma 8.*

Proof. Since $u > c$ on V and $a > c$, the compact slab M is contained in the interior of V. On M the vector field

$$X := \frac{\nabla u}{|\nabla u|^2}$$

is smooth and satisfies $Xu = 1$.

If $z \in V_{\geq a}$ and $u(z) < b$, then along the trajectory of X through z the value of u increases linearly:

$$u(\varphi_\tau(z)) = u(z) + \tau.$$

Hence the trajectory is defined for all $0 \leq \tau \leq b - u(z)$ and remains in $M \subset V$. This proves that $R_{a,b}$ is well defined. Define

$$H : [0,1] \times V_{\geq a} \to V_{\geq a}$$

by

$$H(s,z) = \begin{cases} z, & u(z) \geq b, \\ \varphi_{s(b-u(z))}(z), & a \leq u(z) < b. \end{cases}$$

If $z_0 \in V_{\geq a}$ and $u(z_0) < b$, then the trajectory segment

$$\{\varphi_\tau(z_0) : 0 \leq \tau \leq b - u(z_0)\}$$

lies in the compact slab M, on which X is smooth. Standard continuous dependence for smooth ODEs therefore implies that H is continuous near each point of $[0,1] \times (V_{\geq a} \setminus V_{\geq b})$. On $V_{\geq b}$ we have $H(s,z) = z$, and at points with $u(z) = b$ the two formulas agree. Therefore H is continuous on all of $[0,1] \times V_{\geq a}$. It is immediate that

$$H(0,z) = z, \qquad H(1,z) = R_{a,b}(z), \qquad H(s,z) = z \text{ for } z \in V_{\geq b}.$$

Thus $R_{a,b}$ is a strong deformation retraction. This proves (a).

For (b), let $z \in V_{>a}$. If $u(z) \geq b$, then $z \in V_{\geq b}$. Otherwise the trajectory segment

$$\tau \mapsto \varphi_\tau(z), \qquad 0 \leq \tau \leq b - u(z),$$

lies in $V_{>a}$ and ends at $R_{a,b}(z) \in V_{\geq b}$. Therefore every point of $V_{>a}$ lies in the same connected component of $V_{>a}$ as some point of $V_{\geq b}$. If $V_{\geq b}$ is connected, it follows that $V_{>a}$ is connected.

For (c), let S be a connected component of $V \cap \{a < u < b\}$. Since $\partial V \subset \{u = c\}$ and $c < a$, every boundary point of S lying in V belongs to one of the levels $\{u = a\}$ or $\{u = b\}$. Thus

$$\partial S \setminus \Big(\{u = a\} \cup \{u = b\}\Big) = \varnothing,$$

so the side-boundary hypothesis of Lemma 8 is vacuous. Applying Lemma 8 to S gives the desired strip structure. □

Component inheritance for nested superlevel sets. We shall use the following elementary observation repeatedly. Let V be a connected component of

$$\{u > c\},$$

and let $d > c$. If C is a connected component of

$$V \cap \{u > d\},$$

then C is also a connected component of the global superlevel set $\{u > d\}$. Indeed, if D is the connected component of $\{u > d\}$ containing C, then $D \subset \{u > c\}$. Since D is connected and meets the connected component V of $\{u > c\}$, we must have $D \subset V$. Hence $D \subset V \cap \{u > d\}$, and by maximality of C inside $V \cap \{u > d\}$ we get $D = C$.

Lemma 11 (Geometric realization of a finite tree)**.** *Let T be a finite tree with vertex set $V(T)$ and edge set $E(T)$. Suppose that to each vertex $v \in V(T)$ there is assigned a point $p_v \in U$, and to each edge $e = \{v, w\} \in E(T)$ there is assigned an embedded arc*

$$A_e \subset U$$

joining p_v to p_w, such that:

(i) *if $v \neq w$, then $p_v \neq p_w$;*
(ii) *the interior of A_e contains no point $p_{v'}$;*
(iii) *if $e \neq e'$, then the interiors of A_e and $A_{e'}$ are disjoint;*
(iv) *if $e = \{v, w\}$ and $e' = \{v', w'\}$ are distinct, then*

$$A_e \cap A_{e'} \subset \{p_v, p_w\} \cap \{p_{v'}, p_{w'}\}.$$

Equivalently, two distinct edge-arcs meet only at common endpoints, and they meet there exactly when the corresponding edges of T share that vertex.

Then

$$H := \bigcup_{e \in E(T)} A_e$$

is homeomorphic to T.

Proof. For each edge $e = \{v, w\}$ choose a homeomorphism $\varphi_e : [0, 1] \to A_e$ with $\varphi_e(0) = p_v$ and $\varphi_e(1) = p_w$. Because of (ii), no interior point of an edge-arc is identified with a vertex-point. Because of (iii) and (iv), if $e \neq e'$ then points of $[0, 1]$ and $[0, 1]$ can map to the same point of U only when both are endpoints corresponding to a common endpoint-vertex of the two edges. Hence the maps φ_e are compatible with the quotient relation defining the geometric realization $|T|$ of the abstract tree. They therefore glue to a continuous map

$$\Phi : |T| \to H.$$

The map Φ is surjective by construction. It is injective because distinct open edges of $|T|$ map into disjoint interiors by (iii), and an open edge can meet another edge-image only at shared endpoint-vertices by (iv). Thus Φ is a continuous bijection from the compact space $|T|$ onto the Hausdorff space $H \subset \mathbb{C}$, hence a homeomorphism. □

Auxiliary topological fact. Let $\Omega_0, \Omega_1 \subset \mathbb{C}$ be Jordan domains such that

$$\Omega_1 \subset \Omega_0 \qquad \text{and} \qquad \partial\Omega_1 \cap \partial\Omega_0 = \varnothing.$$

Then $\overline{\Omega_1} \subset \Omega_0$, and $\Omega_0 \setminus \overline{\Omega_1}$ is homeomorphic to an annulus.

Indeed, since $\overline{\Omega_1}$ is connected, meets Ω_0, and is disjoint from $\partial\Omega_0$, it cannot meet the exterior component of $\mathbb{C} \setminus \partial\Omega_0$. Hence $\overline{\Omega_1} \subset \Omega_0$. Now choose, by the Jordan–Schoenflies theorem, a homeomorphism

$$h : \overline{\Omega_0} \to \overline{\mathbb{D}}.$$

Then $h(\partial\Omega_1)$ is a Jordan curve contained in $\mathbb{D}$, so it bounds a Jordan domain $\Delta \Subset \mathbb{D}$. The set $h(\Omega_1)$ is a connected component of $\mathbb{D} \setminus h(\partial\Omega_1)$, whose two components are Δ and $\mathbb{D} \setminus \overline{\Delta}$. Since $h(\overline{\Omega_1})$ is compactly contained in $\mathbb{D}$, its closure does not meet $\partial\mathbb{D}$, so $h(\Omega_1)$ cannot be the outer component $\mathbb{D} \setminus \overline{\Delta}$. Hence $h(\Omega_1) = \Delta$. Therefore

$$h\Big(\Omega_0 \setminus \overline{\Omega_1}\Big) = \mathbb{D} \setminus \overline{\Delta},$$

which is an annulus.

Proposition 12. *Let $c \geq 0$, and let V be a connected component of $\{u > c\}$ containing exactly m zeros of f. Assume that:*

(a) *u has no critical values on V in $[c, 2\alpha]$;*
(b) *all critical points of u in V with value $> 2\alpha$ are nondegenerate;*
(c) *all critical values of u in V larger than 2α are pairwise distinct.*

Then for every $\varepsilon > 0$ there exists a finite embedded tree $G_\varepsilon \subset V$ containing all m zeros of f in V and satisfying

$$\operatorname{len}(G_\varepsilon) \leq \frac{1}{2\pi} \int_{2\alpha}^{\infty} P_V(t)\, dt + \varepsilon, \tag{10}$$

where $P_V(t) := \mathcal{H}^1(\{z \in V : u(z) = t\})$ for regular $t > 2\alpha$.

Proof. If $m = 1$, then the one-point set consisting of the unique zero has length 0, so the conclusion is immediate. Thus we may assume that $m \geq 2$.

Let the distinct critical values of u in V above 2α be

$$2\alpha < \mu_1 < \mu_2 < \cdots < \mu_s.$$

We claim that $s \geq 1$.

Indeed, suppose for contradiction that $s = 0$. Then there are no critical values of u in $(2\alpha, \infty)$. We first claim that

$$V_{\geq 2\alpha} := \{z \in V : u(z) \geq 2\alpha\}$$

is connected.

Let $z \in V$ with $u(z) < 2\alpha$, and write $s_z := u(z) \in (c, 2\alpha)$. Set

$$M_z := V \cap \{s_z \leq u \leq 2\alpha\}.$$

Because V is a connected component of the open set $\{u > c\}$, it is relatively closed there; since $s_z > c$, the set M_z is closed in $\mathbb{C}$. It is also bounded because Λ_f is bounded, hence compact. As u has no critical points on M_z, the vector field

$$X = \frac{\nabla u}{|\nabla u|^2}$$

is smooth on a neighborhood of M_z. Along the maximal trajectory $\phi_\tau(z)$ through z one has

$$u(\phi_\tau(z)) = s_z + \tau,$$

so the trajectory stays in M_z for $0 \le \tau \le 2\alpha - s_z$ and therefore exists on that whole interval. Thus every point of V flows inside V to a point of $V_{\ge 2\alpha}$. Defining the homotopy by flowing each point upward until it reaches level 2α (and then keeping it fixed), one obtains, by the same continuity argument used in Lemma 10(a) on each compact subslab $V \cap \{s \le u \le 2\alpha\}$ with $c < s < 2\alpha$, a strong deformation retraction of V onto $V_{\ge 2\alpha}$. Since V is connected, $V_{\ge 2\alpha}$ is connected.

Now let $T > 2\alpha$ be a sufficiently large regular value such that $\{z \in V : u(z) > T\}$ has exactly m connected components, one near each zero of f in V. Since there are no critical values of u in $[2\alpha, T]$, Lemma 10(a) gives a strong deformation retraction of $V_{\ge 2\alpha}$ onto

$$V_{\ge T} := \{z \in V : u(z) \ge T\}.$$

Hence $V_{\ge T}$ is connected.

Let $W_1, \dots, W_m$ be the connected components of $\{z \in V : u(z) > T\}$. Since T is a regular value of u, every point $p \in \{z \in V : u(z) = T\}$ has a neighborhood $U_p \subset V$ in which, after a smooth change of coordinates, $u - T = x$. Hence

$$U_p \cap \{u > T\}$$

is connected. It follows that at most one component W_j can accumulate at p. Therefore the closures $\overline{W_1}^V, \dots, \overline{W_m}^V$ are pairwise disjoint. Moreover,

$$V_{\ge T} = \bigcup_{j=1}^{m} \overline{W_j}^V.$$

Since the closure of a connected set is connected, each $\overline{W_j}^V$ is connected. Thus these m sets are precisely the connected components of $V_{\ge T}$. Because $m \ge 2$, this contradicts the connectedness of $V_{\ge T}$.

This contradiction shows that $s \ge 1$.

Choose pairwise disjoint closed intervals

$$I_r = [\mu_r - \delta_r, \mu_r + \delta_r] \qquad (1 \le r \le s)$$

so small that each I_r contains no critical value of u other than μ_r. Choose a regular value

$$\tau_0 \in (2\alpha, \mu_1 - \delta_1),$$

and choose a regular value

$$T > \mu_s + \delta_s$$

so large that the connected components

$$Z_1, \dots, Z_m$$

of $\{z \in V : u(z) > T\}$ have pairwise disjoint closures.

For each j set

$$K_j := \overline{Z_j}.$$

Then $K_1, \dots, K_m$ are pairwise disjoint compact sets, each lying in a small neighborhood of one zero a_j of f. Since f is squarefree, each Z_j (and hence each K_j) contains exactly one zero a_j.

We first note that $\{z \in V : u > \tau_0\}$ is connected. Choose a regular value $t_0 \in (c, 2\alpha)$. As above, the absence of critical points on $V \cap \{c < u \le t_0\}$ gives a strong deformation retraction of V onto

$$V_{\ge t_0} := \{z \in V : u \ge t_0\},$$

so $V_{\ge t_0}$ is connected. Since there are no critical points in $\{z \in V : t_0 \le u \le \tau_0\}$, Lemma 10(a) shows that

$$V_{\ge \tau_0} := \{z \in V : u \ge \tau_0\}$$

is connected. Next choose a regular value $\tau_1 \in (\tau_0, \mu_1 - \delta_1)$. Applying Lemma 10(a) on $[\tau_0, \tau_1]$, we find that $V_{\ge \tau_1}$ is connected. Lemma 10(b) applied with $a = \tau_0$ and $b = \tau_1$ now gives that $\{z \in V : u > \tau_0\}$ is connected.

For each $r = 1, \dots, s$, let p_r be the unique critical point of u with critical value μ_r. Since p_r is nondegenerate and u is harmonic, the Morse normal form gives local coordinates (x, y) centered at p_r in which

$$u = \mu_r + x^2 - y^2;$$

see, for example, [4]. Choose pairwise disjoint closed neighborhoods N_r of p_r so small that:

(i) $N_r \subset u^{-1}(I_r)$;

(ii) ∂N_r is the union of one connected incoming cross-section

$$C_r^- \subset \{u = \mu_r - \delta_r\},$$

two connected outgoing cross-sections

$$C_{r,1}^+,\ C_{r,2}^+ \subset \{u = \mu_r + \delta_r\},$$

and finitely many flow segments of X joining the endpoints of these cross-sections;

(iii) inside N_r, the local superlevel picture is the standard saddle picture: crossing the level μ_r upward changes one component into two components;

(iv) outside N_r, there are no critical points in the slab

$$\mu_r - \delta_r \le u \le \mu_r + \delta_r.$$

Set

$$\Omega := \{z \in V : \tau_0 < u < T\} \setminus \bigcup_{r=1}^{s} N_r.$$

We shall prove below that, after shrinking the neighborhoods N_r further if necessary, each of the cross-sections $C_r^-, C_{r,1}^+, C_{r,2}^+$ lies on the boundary of a unique connected component of Ω.

Let S be a connected component of Ω. Since u has no critical points on S and every side-boundary component of S is a flow line of X, Lemma 8 applies: S is a regular strip, with one connected lower cross-section and one connected upper cross-section, and every level section

$$\Gamma_t := S \cap \{u = t\}$$

is connected.

For such a strip S, let $a_S < b_S$ be the levels of its lower and upper cross-sections. For $t \in (a_S, b_S)$, let $C_t(S)$ denote the connected component of

$$\{z \in V : u(z) > t\}$$

that meets $S \cap \{t < u < t + \eta\}$ for some (equivalently, every sufficiently small) $\eta > 0$. Let k_S be the number of zeros of f in $C_t(S)$. This is independent of $t \in (a_S, b_S)$. Indeed, if $t_1 < t_2$ both lie in (a_S, b_S), then the slab

$$M := C_{t_1}(S) \cap \{t_1 \leq u \leq t_2\}$$

contains no critical points of u. Hence the flow of

$$X = \frac{\nabla u}{|\nabla u|^2}$$

provides the regular continuation of the component determined by S from level t_1 up to level t_2; in particular, no splitting or merging of this component can occur between these two regular levels. Thus the component of $\{u > t_2\}$ obtained from $C_{t_1}(S)$ by passing through the slab is exactly $C_{t_2}(S)$. Since every zero of f corresponds to the singular level $u = +\infty$, no zero lies in the finite slab $\{t_1 \leq u \leq t_2\}$, and therefore the set of zeros contained in the continued component is the same at levels t_1 and t_2. Hence the number of zeros in $C_t(S)$ is constant on (a_S, b_S).

Set

$$\Gamma_t := S \cap \{u = t\}.$$

We claim that

$$\partial C_t(S) = \Gamma_t.$$

Since t is a regular value and $C_t(S)$ is a connected component of $\{u > t\} \cap V$, Lemma 9 implies that $\partial C_t(S)$ is a smooth Jordan curve contained in $\{u = t\} \cap V$.

Because $t \in (a_S, b_S) \subset (\tau_0, T) \setminus \bigcup_{r=1}^{s} I_r$, the level $\{u = t\} \cap V$ does not meet any N_r. Hence

$$\{u = t\} \cap V = \{u = t\} \cap \Omega,$$

and Γ_t is a connected component of $\Omega \cap \{u = t\}$.

First, $\Gamma_t \subset \partial C_t(S)$: if $x \in \Gamma_t$, then for every sufficiently small $\eta > 0$ the set $S \cap \{t < u < t + \eta\}$ lies in $C_t(S)$ by definition, and its points converge to x. Since $x \notin C_t(S)$, this shows $x \in \partial C_t(S)$.

Conversely, $\partial C_t(S)$ is a connected subset of $\Omega \cap \{u = t\}$, and it meets Γ_t by the previous paragraph. Because Γ_t is a connected component of $\Omega \cap \{u = t\}$, it follows that

$$\partial C_t(S) \subset \Gamma_t.$$

Combining the two inclusions gives

$$\partial C_t(S) = \Gamma_t.$$

Applying the auxiliary flux lemma to the component $C_t(S)$ gives

$$\int_{\Gamma_t} |\nabla u| \, ds = 2\pi k_S.$$

For an arc $I \subset \Gamma_{t_1}$, let I_{t_2} be its image under the flow from level t_1 to level t_2. The region swept by these trajectories has boundary consisting of I, I_{t_2}, and two flow lines.

Since $\Delta u = 0$ in this region and ∇u is tangent to the side flow lines, Green's formula gives

$$\int_I |\nabla u|\, ds = \int_{I_{t_2}} |\nabla u|\, ds.$$

Thus the measures

$$d\mu_t := \frac{|\nabla u|\, ds}{2\pi k_S}$$

on Γ_t are probability measures preserved by the flow.

Fix $s_0 \in (a_S, b_S)$. For $x \in \Gamma_{s_0}$, let

$$\widetilde{\gamma}_x(t) := \Phi_{s_0}(x,t), \qquad t \in (a_S, b_S),$$

where Φ_{s_0} is the strip diffeomorphism from Lemma 8. We claim that $\widetilde{\gamma}_x$ extends continuously to $[a_S, b_S]$, with endpoint values on the lower and upper cross-sections of S. Indeed, let K_x^- and K_x^+ be the closures in $\mathbb{C}$ of the sets

$$S \cap \{a_S < u < s_0\}, \qquad S \cap \{s_0 < u < b_S\},$$

respectively. These sets are compact because

$$S \cap \{a_S < u < s_0\} \subset \{a_S \le u \le s_0\}, \qquad S \cap \{s_0 < u < b_S\} \subset \{s_0 \le u \le b_S\},$$

and the right-hand sides are closed and bounded subsets of $\mathbb{C}$. Moreover every point of $K_x^\pm$ is noncritical: interior points because S contains no critical points, points on the lower or upper cross-sections because these lie on regular levels of u, and points on the side boundary because those components are integral curves of X. All such points lie in the interior of $V \setminus \{p_1, \dots, p_s\}$, so by continuity X is smooth on a neighborhood of $K_x^- \cup K_x^+$. Since $Xu = 1$, standard ODE continuation shows that the trajectory through x extends to times $a_S - s_0$ and $b_S - s_0$. Thus

$$\gamma_x := \widetilde{\gamma}_x([a_S, b_S])$$

is a well-defined compact trajectory segment joining the lower cross-section of S to its upper cross-section. For $x \in \Gamma_{s_0}$, define

$$\ell(x) := \operatorname{len}(\gamma_x).$$

Since $\widetilde{\gamma}_x(t) = \Phi_{s_0}(x,t)$ for $t \in [a_S, b_S]$, we have

$$\ell(x) = \int_{a_S}^{b_S} |\partial_t \Phi_{s_0}(x,t)|\, dt = \int_{a_S}^{b_S} \left| X\Big(\Phi_{s_0}(x,t)\Big) \right|\, dt.$$

Because

$$X = \frac{\nabla u}{|\nabla u|^2},$$

it follows that

$$|X| = \frac{1}{|\nabla u|},$$

and hence

$$\ell(x) = \int_{a_S}^{b_S} \frac{dt}{|\nabla u(\Phi_{s_0}(x,t))|}.$$

Integrating with respect to the probability measure μ_{s_0} and using Fubini,

$$\int_{\Gamma_{s_0}} \ell(x)\, d\mu_{s_0}(x) = \int_{a_S}^{b_S} \left(\int_{\Gamma_{s_0}} \frac{1}{|\nabla u(\Phi_{s_0}(x,t))|}\, d\mu_{s_0}(x) \right) dt.$$

Since the flow preserves the measures μ_t, we have

$$(\Phi_{s_0}(\cdot,t))_{\#}\mu_{s_0} = \mu_t.$$

Therefore the inner integral equals

$$\int_{\Gamma_t} \frac{1}{|\nabla u(y)|}\, d\mu_t(y) = \frac{1}{2\pi k_S}\int_{\Gamma_t} ds = \frac{P_S(t)}{2\pi k_S},$$

where $P_S(t) := \mathcal{H}^1(\Gamma_t)$. Thus

$$\int_{\Gamma_{s_0}} \operatorname{len}(\gamma_x)\, d\mu_{s_0}(x) = \frac{1}{2\pi k_S}\int_{a_S}^{b_S} P_S(t)\, dt.$$

Since μ_{s_0} is a probability measure, there exists $x \in \Gamma_{s_0}$ such that

$$\operatorname{len}(\gamma_x) \le \frac{1}{2\pi k_S}\int_{a_S}^{b_S} P_S(t)\, dt.$$

Fix such an x and denote the corresponding trajectory segment by γ_S.

Summing over all strips and using $k_S \ge 1$, we obtain

$$\sum_S \operatorname{len}(\gamma_S) \le \frac{1}{2\pi}\sum_S \int_{a_S}^{b_S} P_S(t)\, dt.$$

Since the strips are disjoint and

$$\sum_S P_S(t) = \mathcal{H}^1(\{u = t\} \cap \Omega) \le P_V(t)$$

for every regular $t \in (\tau_0, T)$, Fubini's theorem yields

$$\sum_S \operatorname{len}(\gamma_S) \le \frac{1}{2\pi}\int_{\tau_0}^{T} P_V(t)\, dt \le \frac{1}{2\pi}\int_{2\alpha}^{\infty} P_V(t)\, dt. \tag{11}$$

We now prove the uniqueness statements needed for the combinatorics.

Bottom strip. Set

$$W_{\tau_0} := V \cap \{u > \tau_0\}.$$

We already know that W_{τ_0} is connected. Since there are no critical points on

$$V \cap \{\tau_0 \le u \le \mu_1 - \delta_1\},$$

Lemma 10(a), applied on $[\tau_0, \mu_1 - \delta_1]$, shows that

$$V^{\ge \mu_1 - \delta_1} := V \cap \{u \ge \mu_1 - \delta_1\}$$

is connected because $V^{\ge \tau_0}$ is connected. Choose a regular value

$$\tau_1 \in (\mu_1 - \delta_1, \mu_1).$$

Applying Lemma 10(a) on $[\mu_1 - \delta_1, \tau_1]$ shows that $V^{\ge \tau_1}$ is connected. Then Lemma 10(b), applied with $a = \mu_1 - \delta_1$ and $b = \tau_1$, implies that

$$W_1 := V \cap \{u > \mu_1 - \delta_1\}$$

is connected.

By the component-inheritance paragraph above, both W_{τ_0} and W_1 are connected components of the global regular superlevel sets $\{u > \tau_0\}$ and $\{u > \mu_1 - \delta_1\}$, respectively. Hence Lemma 9 applies to both of them, so each is a Jordan domain. Since

$$\partial W_{\tau_0} \subset \{u = \tau_0\}, \qquad \partial W_1 \subset \{u = \mu_1 - \delta_1\},$$

these boundaries are disjoint. Because $W_1 \subset W_{\tau_0}$, the auxiliary topological fact implies that

$$\overline{W_1} \subset W_{\tau_0}$$

and that

$$W_{\tau_0} \setminus \overline{W_1}$$

is an annulus. Since $\partial W_1 \subset \{u = \mu_1 - \delta_1\}$, we have

$$A_0 := V \cap \{\tau_0 < u < \mu_1 - \delta_1\} = W_{\tau_0} \setminus \overline{W_1}.$$

Therefore A_0 is a connected annulus. Since there are no critical points on A_0, Lemma 10(c) implies that every connected component of A_0 is a regular strip. Because A_0 is connected, it is a single regular strip. Hence there is exactly one strip whose lower cross-section lies in $\{u = \tau_0\}$; denote it by S_0.

Critical neighborhoods. Claim. After shrinking the pairwise disjoint neighborhoods N_r further if necessary, each of the cross-sections C_r^-, $C_{r,1}^+$, and $C_{r,2}^+$ lies on the boundary of exactly one connected component of Ω. More precisely, there exist unique connected components

$$S_r^-, \qquad S_{r,1}^+, \qquad S_{r,2}^+$$

of Ω such that C_r^- is the upper cross-section of S_r^- and $C_{r,1}^+, C_{r,2}^+$ are the lower cross-sections of $S_{r,1}^+, S_{r,2}^+$, respectively.

Proof of claim. Fix r. Choose Morse coordinates (x, y) on a neighborhood U_r of p_r such that

$$u = \mu_r + x^2 - y^2,$$

and shrink $N_r \Subset U_r$ so that the three cross-sections

$$C_r^- \subset \{u = \mu_r - \delta_r\}, \qquad C_{r,1}^+, C_{r,2}^+ \subset \{u = \mu_r + \delta_r\}$$

appear exactly as in the standard saddle picture, with the remainder of ∂N_r consisting of flow segments of X.

In these coordinates, each of the three cross-sections carries a one-sided collar in $U_r \setminus N_r$ contained in Ω: the collar adjacent to C_r^- lies on the lower side of the saddle, while the collars adjacent to $C_{r,1}^+$ and $C_{r,2}^+$ lie on the two upper sides. Each such collar is connected. Therefore each cross-section is contained in the boundary of exactly one connected component of Ω; denote these components by

$$S_r^-, \qquad S_{r,1}^+, \qquad S_{r,2}^+.$$

The same local coordinate picture shows that C_r^- is the upper cross-section of S_r^-, while $C_{r,1}^+$ and $C_{r,2}^+$ are the lower cross-sections of $S_{r,1}^+$ and $S_{r,2}^+$, respectively. This proves the claim. $\triangle$ *Top strips.* Fix $j \in \{1, \dots, m\}$ and set

$$c_j := \mu_s + \delta_s.$$

Let W_j be the connected component of

$$V \cap \{u > c_j\}$$

that contains K_j, and set

$$Y_j := W_j \cap \{u \geq T\}.$$

By the component-inheritance paragraph above, W_j is also a connected component of the global regular superlevel set $\{u > c_j\}$. Hence Lemma 9 implies that W_j is a Jordan domain.

We claim that Y_j is connected. Define

$$R_j(z) := \begin{cases} z, & u(z) \geq T, \\ \varphi_{T-u(z)}(z), & c_j < u(z) < T, \end{cases}$$

and

$$H_j(\theta, z) := \begin{cases} z, & u(z) \geq T, \\ \varphi_{\theta(T-u(z))}(z), & c_j < u(z) < T. \end{cases}$$

If $z_0 \in W_j \setminus Y_j$, then the trajectory segment

$$\{\varphi_\tau(z_0) : 0 \leq \tau \leq T - u(z_0)\}$$

stays in the compact set

$$W_j \cap \{u(z_0) \leq u \leq T\},$$

on which X is smooth because there are no critical points on $W_j \cap \{c_j < u \leq T\}$. Standard continuous dependence for smooth ODEs therefore shows that H_j is continuous on $[0,1] \times W_j$. Thus H_j is a strong deformation retraction of W_j onto Y_j, and Y_j is connected.

Since T is a regular value, the connected components of $V \cap \{u \geq T\}$ are precisely $K_1, \ldots, K_m$. The set $Y_j \subset V \cap \{u \geq T\}$ is connected and contains K_j, hence $Y_j \subset K_j$. The reverse inclusion is immediate, so $Y_j = K_j$. In particular Z_j is a Jordan domain and $K_j = \overline{Z_j}$. Since

$$\partial W_j \subset \{u = c_j\}, \qquad \partial Z_j = \partial K_j \subset \{u = T\},$$

these boundaries are disjoint, and $Z_j \subset W_j$. Hence, by the auxiliary topological fact,

$$K_j = \overline{Z_j} \subset W_j$$

and

$$W_j \setminus K_j = W_j \setminus \overline{Z_j}$$

is an annulus. Because $\partial K_j \subset \{u = T\}$, we obtain

$$A_j := W_j \cap \{c_j < u < T\} = W_j \setminus K_j.$$

Therefore A_j is a connected annulus. Since there are no critical points on A_j, Lemma 10(c) implies that every connected component of A_j is a regular strip. Because A_j is connected, it is a single regular strip. Hence there is exactly one strip of Ω whose upper cross-section is ∂K_j; denote it by S_j^{top}.

Exhaustion lemma for the strips. Every connected component S of Ω has its lower cross-section in exactly one of

$$\{u = \tau_0\}, \qquad C_{r,1}^+, C_{r,2}^+ \quad (1 \leq r \leq s),$$

and its upper cross-section in exactly one of

$$C_r^- \quad (1 \leq r \leq s), \qquad \partial K_j \quad (1 \leq j \leq m).$$

Conversely, the unique strip with lower cross-section in $\{u = \tau_0\}$ is S_0; the unique strips with lower cross-sections $C^+_{r,1}, C^+_{r,2}$ are $S^+_{r,1}, S^+_{r,2}$; the unique strips with upper cross-sections C^-_r are S^-_r; and the unique strips with upper cross-sections ∂K_j are S^{top}_j.

Proof. Let S be a connected component of Ω. By Lemma 8, S is a regular strip, so it has exactly one lower cross-section and exactly one upper cross-section, both lying on boundary components of Ω contained in level sets of u. Now $\partial\Omega$ consists of:

- the base level $\{u = \tau_0\}$,
- the top boundaries $\partial K_j \subset \{u = T\}$,
- the cross-sections $C^-_r \subset \{u = \mu_r - \delta_r\}$ and $C^+_{r,1}, C^+_{r,2} \subset \{u = \mu_r + \delta_r\}$,
- and side-boundary flow segments on the sets ∂N_r.

By definition, a cross-section of a regular strip cannot lie on a side-boundary flow segment. Since the lower level of S is strictly smaller than its upper level, the lower cross-section cannot lie on any ∂K_j or C^-_r, and the upper cross-section cannot lie on $\{u = \tau_0\}$ or any $C^+_{r,i}$. This proves the classification.

The converse uniqueness statements are exactly the uniqueness assertions for bottom strips, critical neighborhoods, and top strips proved above. △

We now define a finite abstract graph T. Its vertices are

$$V(T) = \{v_0\} \cup \{w_1, \dots, w_s\} \cup \{\ell_1, \dots, \ell_m\},$$

where v_0 is the root, w_r corresponds to the saddle p_r, and ℓ_j corresponds to the zero a_j.

For each strip $S \subset \Omega$, define an oriented edge e_S from the vertex attached to its lower cross-section to the vertex attached to its upper cross-section:

- if the lower cross-section of S lies in $\{u = \tau_0\}$, then the lower endpoint of e_S is v_0;
- if the lower cross-section of S is one of the arcs $C^+_{r,i}$, then the lower endpoint of e_S is w_r;
- if the upper cross-section of S is C^-_r, then the upper endpoint of e_S is w_r;
- if the upper cross-section of S is ∂K_j, then the upper endpoint of e_S is ℓ_j.

By the exhaustion lemma above, every strip determines exactly one such edge.

Every non-root vertex has exactly one incoming edge: w_r receives the edge $e_{S^-_r}$, and ℓ_j receives the edge $e_{S^{\mathrm{top}}_j}$. Likewise each w_r has exactly two outgoing edges, namely $e_{S^+_{r,1}}$ and $e_{S^+_{r,2}}$.

Assign heights by

$$h(v_0) = \tau_0, \qquad h(w_r) = \mu_r, \qquad h(\ell_j) = T + 1.$$

Every oriented edge goes from a vertex of smaller height to a vertex of larger height. Hence there is no directed cycle. Moreover, iterating the unique predecessor map from any non-root vertex strictly decreases height, so after finitely many steps one reaches v_0. Thus every vertex is connected to v_0, and the graph is connected and acyclic. Therefore T is a finite rooted tree.

We now realize T geometrically. The neighborhoods N_r and the value T have already been fixed, and hence so have Ω, its strips S, and the trajectory segments γ_S.

For each r, let

$$x_{r,-} := \gamma_{S^-_r} \cap C^-_r, \qquad x_{r,i} := \gamma_{S^+_{r,i}} \cap C^+_{r,i} \quad (i = 1, 2).$$

These points are well defined, because S_r^- has upper cross-section C_r^- and $S_{r,i}^+$ has lower cross-section $C_{r,i}^+$. They are pairwise distinct, since the three cross-sections

$$C_r^-,\ C_{r,1}^+,\ C_{r,2}^+$$

are disjoint subsets of ∂N_r. Choose a point $c_r \in \mathrm{int}(N_r)$. Because N_r is a closed topological disk, there exists an embedded Y-shaped tree

$$T_r \subset N_r$$

with branch point c_r and three terminal vertices $x_{r,-}, x_{r,1}, x_{r,2}$. Since the neighborhoods N_r may be chosen arbitrarily small at the outset and are pairwise disjoint, we may assume in addition that

$$\sum_{r=1}^{s} \mathrm{len}(T_r) < \frac{\varepsilon}{2}. \tag{12}$$

Let x_0 be the lower endpoint of γ_{S_0} on the level $\{u = \tau_0\}$. For each j, let b_j be the upper endpoint of $\gamma_{S_j^{\mathrm{top}}}$ on ∂K_j. Write

$$f(z) = (z - a_j)^{m_j} h_j(z)$$

in a neighborhood of a_j, where $m_j \geq 1$ and h_j is holomorphic with $h_j(a_j) \neq 0$. After shrinking to a sufficiently small closed disk $B(a_j, r_j)$, choose a holomorphic branch q_j of h_j^{1/m_j} there and define

$$\psi_j(z) := (z - a_j) q_j(z).$$

Then ψ_j is biholomorphic on a neighborhood of $B(a_j, r_j)$, one has $\psi_j(a_j) = 0$, and

$$f(z) = \psi_j(z)^{m_j} \qquad (z \in B(a_j, r_j)).$$

By the choice of T, each K_j is contained in $B(a_j, r_j)$ and

$$Z_j = \psi_j^{-1}\Big(D(0, e^{-T/m_j})\Big), \qquad K_j = \psi_j^{-1}\Big(\overline{D(0, e^{-T/m_j})}\Big).$$

If $w_j := \psi_j(b_j)$, then $|w_j| = e^{-T/m_j}$, so the radial segment $[0, w_j]$ is contained in $\overline{D(0, e^{-T/m_j})}$ and its relative interior is contained in $D(0, e^{-T/m_j})$. Therefore its preimage under ψ_j^{-1} is an embedded arc

$$\sigma_j \subset K_j, \qquad \sigma_j^\circ \subset Z_j,$$

joining a_j to b_j. Since ψ_j^{-1} is C^1 near 0, there exists $C_j > 0$ such that

$$\mathrm{len}(\sigma_j) \leq C_j e^{-T/m_j}.$$

Hence, by the choice of T,

$$\sum_{j=1}^{m} \mathrm{len}(\sigma_j) < \frac{\varepsilon}{2}. \tag{13}$$

For each strip S, define an embedded arc A_S as follows:

- if e_S joins v_0 to w_r, let A_S be the union of γ_S and the unique arm of T_r joining the upper endpoint of γ_S to c_r;
- if e_S joins w_r to $w_{r'}$, let A_S be the union of the unique arm of T_r from c_r to the lower endpoint of γ_S, the segment γ_S, and the unique arm of $T_{r'}$ from the upper endpoint of γ_S to $c_{r'}$;

- if e_S joins w_r to ℓ_j, let A_S be the union of the unique arm of T_r from c_r to the lower endpoint of γ_S, the segment γ_S, and the arc σ_j from the upper endpoint of γ_S to a_j.

The geometric vertices are

$$p_{v_0} = x_0, \qquad p_{w_r} = c_r, \qquad p_{\ell_j} = a_j.$$

We verify the hypotheses of Lemma 11 for the family of arcs $\{A_S\}$. First, if $S \neq S'$, then

$$\gamma_S \cap \gamma_{S'} = \varnothing,$$

because distinct strips are disjoint subsets of Ω. Second, for each r, the tree $T_r \subset N_r$ meets the rest of the construction only at its terminal points $x_{r,-}, x_{r,1}, x_{r,2}$: indeed the neighborhoods N_r are pairwise disjoint, each N_r is disjoint from every Z_j, and a strip trajectory can meet ∂N_r only at the cross-section belonging to that strip. Third, for each j, the arc $\sigma_j \subset K_j \subset Z_j$ meets the rest of the construction only at its endpoint $b_j \in \partial K_j$. Indeed, the sets $Z_1, \dots, Z_m$ are pairwise disjoint and disjoint from all N_r, so the arcs σ_j are pairwise disjoint and are disjoint from every T_r.

It follows that the interior of each A_S contains no geometric vertex, the interiors of distinct arcs A_S and $A_{S'}$ are disjoint, and two distinct arcs can meet only at a common geometric endpoint corresponding to a common endpoint of the associated edges of T. Hence Lemma 11 applies:

$$H := \bigcup_S A_S$$

is homeomorphic to T. Since T is a tree, H is a finite embedded tree in V containing all zeros $a_1, \dots, a_m$.

Finally,

$$\operatorname{len}(H) \le \sum_S \operatorname{len}(\gamma_S) + \sum_{r=1}^{s} \operatorname{len}(T_r) + \sum_{j=1}^{m} \operatorname{len}(\sigma_j).$$

Using (11), (12), and (13), we obtain

$$\operatorname{len}(H) \le \frac{1}{2\pi} \int_{2\alpha}^{\infty} P_V(t)\, dt + \varepsilon.$$

Setting $G_\varepsilon := H$ proves (10). □

5. An explicit perturbation lemma

We now replace the abstract transversality step by a concrete two-parameter perturbation. Fix a squarefree monic polynomial f, a component $U \subset \Lambda_f$ containing exactly $m \ge 2$ zeros of f, and the number α introduced above. Define the compact collars

$$K := \{z \in U : u(z) \ge \alpha/2\}, \qquad L := \{z \in U : u(z) \ge 3\alpha/2\}.$$

Then K and L are compact and connected by Lemma 7.

Set

$$c := 3\alpha/2.$$

Since u has no critical values in $[0, 4\alpha)$, the value c is regular. Let

$$W_1, \dots, W_r$$

be the connected components of $\{z \in U : u(z) > c\}$. We claim that the closures $\overline{W_i}$ are pairwise disjoint. Indeed, if $p \in \overline{W_i} \cap \overline{W_j}$, then $u(p) = c$. Because $c > 0$, the point p lies in the interior of U. Since c is a regular value, the implicit function theorem gives local coordinates near p in which

$$u - c = x.$$

Hence locally $\{u > c\}$ is a single open half-disk, so only one connected component of $\{z \in U : u(z) > c\}$ can accumulate at p. Thus the sets $\overline{W_i}$ are pairwise disjoint. Every point of

$$L = \{z \in U : u(z) \geq c\}$$

lies in $\bigcup_i \overline{W_i}$: if $u(p) > c$, then $p \in \{u > c\} = \bigcup_i W_i \subset \bigcup_i \overline{W_i}$; if $u(p) = c$, then the same local normal form shows that points with $u > c$ lie arbitrarily close to p, so again $p \in \bigcup_i \overline{W_i}$. Therefore

$$L = \bigcup_{i=1}^{r} \overline{W_i}.$$

Since L is connected by Lemma 7 and the $\overline{W_i}$ are pairwise disjoint closed subsets of L, it follows that $r = 1$. Let $W := W_1$. Because $W \subset U \subset \{u > 0\}$ and $c > 0$, the set W is also a connected component of the global superlevel set $\{u > c\}$. Hence Lemma 9 implies that W is a Jordan domain and

$$L = \overline{W}.$$

An entirely analogous argument with

$$d := \alpha/2$$

shows that K is a compact Jordan domain. Indeed, d is also a regular value because u has no critical values in $[0, 4\alpha)$. If $Y_1, \ldots, Y_s$ are the connected components of $\{u > d\}$, then exactly as above their closures are pairwise disjoint and

$$K = \{u \geq d\} = \bigcup_{i=1}^{s} \overline{Y_i}.$$

Since K is connected by Lemma 7, we must have $s = 1$. Let $Y := Y_1$. Because $d > 0$ and $Y \subset U \subset \{u > 0\}$, this set Y is a connected component of the global superlevel set $\{u > d\}$. Hence Lemma 9 implies that Y is a Jordan domain and

$$K = \overline{Y}.$$

Choose pairwise disjoint closed disks

$$D_1, \ldots, D_m \Subset \{u > 5\alpha/2\} \cap W$$

centered at the zeros of f in U, so small that each D_j contains exactly one zero and

$$\delta_D := \min_{z \in \bigcup_{j=1}^{m} D_j} |f'(z)| > 0.$$

In particular,

$$B := \{z \in U : \alpha/2 \leq u(z) \leq 5\alpha/2\}$$

is disjoint from $\bigcup_{j=1}^{m} D_j$. Define

$$L_0 := L \setminus \bigcup_{j=1}^{m} \operatorname{int}(D_j).$$

Since W is a Jordan domain, $\overline{W}$ is homeomorphic to the closed unit disk $\overline{\mathbb{D}}$ by the Jordan–Schoenflies theorem. Under such a homeomorphism, the sets D_j correspond to pairwise disjoint compact topological disks $E_j \subset \mathbb{D}$.

Auxiliary fact. Let Ω be a closed topological disk, and let $E_1, \ldots, E_m \subset \operatorname{int}(\Omega)$ be pairwise disjoint compact topological disks. Then

$$\Omega \setminus \bigcup_{j=1}^{m} \operatorname{int}(E_j)$$

is path connected.

Proof of the auxiliary fact. Set

$$X_r := \Omega \setminus \bigcup_{j=1}^{r} \operatorname{int}(E_j) \qquad (0 \le r \le m),$$

so $X_0 = \Omega$. We first prove by induction on r that each X_r is connected. This is clear for $r = 0$. Assume that X_{r-1} is connected, and consider

$$X_r = X_{r-1} \setminus \operatorname{int}(E_r).$$

Let U be a connected component of $X_r \setminus \partial E_r = X_{r-1} \setminus E_r$. Then U is a nonempty proper open subset of the connected space X_{r-1}, so its boundary in X_{r-1} is nonempty. Every such boundary point must lie in $E_r \cap X_r = \partial E_r$. Hence the closure of each connected component of $X_r \setminus \partial E_r$ in X_r meets ∂E_r.

Now suppose that $X_r = A \cup B$ is a separation. Since ∂E_r is connected, it must lie entirely in one side, say $\partial E_r \subset A$. Each connected component U of $X_r \setminus \partial E_r$ is contained either in A or in B; but if $U \subset B$, then, because B is closed in X_r, the closure of U in X_r would also be contained in B, contradicting $\overline{U} \cap \partial E_r \neq \varnothing$. Thus every such component lies in A, and therefore $B = \varnothing$. So X_r is connected, completing the induction.

Finally, each point of X_m has a neighborhood in X_m homeomorphic either to an open disk or to a closed half-disk. For interior points this is immediate. If x lies on $\partial\Omega$ or on some ∂E_j, then, because the finitely many compact Jordan curves $\partial\Omega, \partial E_1, \ldots, \partial E_m$ are pairwise disjoint, there is a small neighborhood of x meeting no boundary curve other than the one through x. The Jordan–Schoenflies theorem then straightens that boundary curve locally to a line, so the corresponding neighborhood of x in X_m is homeomorphic to a closed half-disk. Thus X_m is locally path connected. Since connected and locally path connected spaces are path connected, X_m is path connected. This proves the auxiliary fact.

Applying this with $\Omega = \overline{\mathbb{D}}$, we obtain that

$$\overline{\mathbb{D}} \setminus \bigcup_{j=1}^{m} \operatorname{int}(E_j)$$

is path connected. Pulling paths back under the Schoenflies homeomorphism shows that

$$L_0 = L \setminus \bigcup_{j=1}^{m} \operatorname{int}(D_j)$$

is path connected, hence connected.

For $\lambda, \beta \in \mathbb{C}$ define

$$g_{\lambda,\beta}(z) := f(z) + \lambda z + \beta, \qquad u_{\lambda,\beta} := -\log |g_{\lambda,\beta}|.$$

Lemma 13 (Explicit perturbation)**.** *There exist arbitrarily small parameters* $(\lambda, \beta) \in \mathbb{C}^2$ *such that, writing* $g := g_{\lambda,\beta}$ *and* $u_g := -\log |g|$*, the following hold:*
(1) *each* D_j *contains exactly one zero of* g*, this zero is simple, and* g *has no zeros on* $K \setminus \bigcup_j D_j$*;*
(2) u_g *has no critical values in* $[\alpha, 2\alpha]$ *on* K*;*
(3) *every critical point of* u_g *in* K *with value* $> 2\alpha$ *is nondegenerate;*
(4) *all critical values of* u_g *in* K *larger than* 2α *are pairwise distinct.*

Proof. We proceed in five steps.

Step 1: keep the zeros in the prescribed disks. Since each D_j contains exactly one simple zero of f and f has no zeros on

$$C := K \setminus \bigcup_{j=1}^{m} \operatorname{int}(D_j),$$

we have

$$\delta_C := \min_{z \in C} |f(z)| > 0.$$

Choose $\eta_0 > 0$ so small that $\eta_0 < \delta_C/2$. If

$$\sup_{z \in K} |g(z) - f(z)| < \eta_0,$$

then g has no zeros on C, since

$$|g(z)| \geq |f(z)| - |g(z) - f(z)| > \delta_C/2 \qquad (z \in C).$$

On the other hand, for each j, since f has exactly one zero in D_j and no zero on ∂D_j, Rouché's theorem on ∂D_j shows that g also has exactly one zero in D_j, counted with multiplicity, provided the same uniform bound holds. Thus each D_j still contains exactly one zero of g, and g has no zeros on

$$K \setminus \bigcup_{j=1}^{m} D_j.$$

Since

$$\sup_{z \in K} |g_{\lambda,\beta}(z) - f(z)| \leq |\lambda| \max_{z \in K} |z| + |\beta|,$$

this zero configuration is preserved for all sufficiently small (λ, β).

Moreover, if

$$|\lambda| < \delta_D/2,$$

then for every $z \in \bigcup_{j=1}^{m} D_j$,

$$|g'(z)| = |f'(z) + \lambda| \geq |f'(z)| - |\lambda| > \delta_D/2,$$

so g' has no zeros on any D_j. Combined with the preceding paragraph, this shows that the unique zero of g in each D_j is simple.

Step 2: make the critical points simple.

On the zero-free set $\{g \neq 0\}$, the function $u_g = -\log|g|$ is smooth, and its critical points are exactly the zeros of g', because

$$\nabla(-\log|g|) = 0 \iff g' = 0 \qquad \text{on } \{g \neq 0\}.$$

But

$$g'(z) = f'(z) + \lambda.$$

A zero ζ of $f' + \lambda$ is multiple if and only if

$$f'(\zeta) + \lambda = 0 \quad \text{and} \quad f''(\zeta) = 0,$$

equivalently,

$$\lambda = -f'(\zeta) \quad \text{with} \quad f''(\zeta) = 0.$$

Therefore the bad set

$$\Sigma_1 := \{-f'(\zeta) : f''(\zeta) = 0\}$$

is finite, and for every $\lambda \notin \Sigma_1$ all zeros of $f' + \lambda$ are simple.

We now show that simplicity of the zero of g' implies nondegeneracy of the corresponding critical point of u_g. Let c satisfy $g(c) \neq 0$, $g'(c) = 0$, and $g''(c) \neq 0$. Choose a simply connected neighborhood of c on which g has no zeros, and let $F = \log g$ be a holomorphic branch there. Then

$$u_g = -\Re F.$$

Since

$$F'(z) = \frac{g'(z)}{g(z)},$$

we have $F'(c) = 0$, and

$$F''(c) = \frac{g''(c)}{g(c)} \neq 0.$$

Hence

$$F(z) = F(c) + \frac{F''(c)}{2}(z-c)^2 + O(|z-c|^3),$$

so

$$u_g(z) = u_g(c) - \Re\left(\frac{F''(c)}{2}(z-c)^2\right) + O(|z-c|^3).$$

The quadratic form $\Re\big(A(z-c)^2\big)$ with $A \neq 0$ is nondegenerate and has signature $(1,1)$. Therefore the Hessian of u_g at c is nondegenerate. Thus every critical point of u_g on $\{g \neq 0\}$ corresponding to a simple zero of g' is a Morse saddle.

Consequently, for every $\lambda \notin \Sigma_1$, every critical point of u_g with finite critical value is nondegenerate.

Step 3: make the complex critical values distinct. Set

$$H_\lambda(z) := f(z) + \lambda z.$$

Its critical points are the zeros of

$$H'_\lambda(z) = f'(z) + \lambda.$$

The critical values of H_λ are contained among the roots of the resultant polynomial

$$R_\lambda(w) := \operatorname{Res}_z\big(f'(z) + \lambda,\ f(z) + \lambda z - w\big).$$

If two distinct critical points of H_λ have the same critical value, then R_λ has a repeated root. Hence the discriminant

$$\Delta(\lambda) := \operatorname{Disc}_w R_\lambda(w)$$

vanishes. Therefore every λ for which the critical values of H_λ are not pairwise distinct belongs to the algebraic set

$$\Sigma_2 := \{\lambda : \Delta(\lambda) = 0\}.$$

To prove that such bad parameters do not fill the plane, it suffices to exhibit one value of λ for which the critical values of H_λ are pairwise distinct. The asymptotic argument below does exactly this: for all sufficiently large $|\lambda|$, the critical points $c_k(\lambda)$ satisfy

$$c_k(\lambda) = \rho\omega_k + O(1),$$

and the corresponding critical values satisfy

$$H_\lambda(c_k(\lambda)) = -(n-1)\rho^n\omega_k + O(|\rho|^{n-1}),$$

so they are pairwise distinct. We now carry out that asymptotic argument in detail. Write

$$f'(z) = nz^{n-1} + Q(z), \qquad \deg Q \le n-2.$$

Fix $R > 0$, to be chosen later. Let $\lambda \neq 0$, choose $\rho \in \mathbb{C}$ such that

$$\rho^{n-1} = -\lambda/n,$$

and let $\omega_1, \dots, \omega_{n-1}$ denote the $(n-1)$st roots of unity. For each k set

$$c_k^0 := \rho\omega_k, \qquad C_k := \{z : |z - c_k^0| = R\}.$$

Then $c_1^0, \dots, c_{n-1}^0$ are precisely the zeros of $nz^{n-1} + \lambda$.

Let $z \in C_k$, so that $z = c_k^0 + \xi$ with $|\xi| = R$. Since

$$nz^{n-1} + \lambda = nz^{n-1} - (c_k^0)^{n-1} = n(z - c_k^0)\sum_{m=0}^{n-2} z^{n-2-m}(c_k^0)^m,$$

we have

$$|nz^{n-1} + \lambda| = nR\left|\sum_{m=0}^{n-2} z^{n-2-m}(c_k^0)^m\right|.$$

Factoring out $(c_k^0)^{n-2}$ gives

$$\sum_{m=0}^{n-2} z^{n-2-m}(c_k^0)^m = (c_k^0)^{n-2}\sum_{m=0}^{n-2}\left(1 + \frac{\xi}{c_k^0}\right)^{n-2-m}.$$

Because R is fixed and $|c_k^0| = |\rho| \to \infty$ as $|\lambda| \to \infty$, the second factor converges uniformly on all the circles C_k to $n-1$. Therefore there exists $\rho_1 > 0$ such that, for all $|\rho| \ge \rho_1$ and all $z \in C_k$,

$$\left|\sum_{m=0}^{n-2} z^{n-2-m}(c_k^0)^m\right| \ge \frac{n-1}{2}|\rho|^{n-2}.$$

Hence

$$|nz^{n-1} + \lambda| \ge \frac{n(n-1)}{2}R|\rho|^{n-2} \qquad (z \in C_k,\ |\rho| \ge \rho_1). \tag{14}$$

Now write

$$Q(z) = \sum_{j=0}^{n-2} a_j z^j$$

and set

$$C_0 := \sum_{j=0}^{n-2} |a_j| \left(\frac{3}{2}\right)^j.$$

If $|\rho| \geq 2R$ and $z \in C_k$, then $|z| \leq |\rho| + R \leq \frac{3}{2}|\rho|$, so

$$|Q(z)| \leq \sum_{j=0}^{n-2} |a_j||z|^j \leq \sum_{j=0}^{n-2} |a_j| \left(\frac{3}{2}\right)^j |\rho|^j \leq C_0|\rho|^{n-2}. \tag{15}$$

Choose $R > 0$ so large that

$$\frac{n(n-1)}{2} R > C_0.$$

Then, by (14) and (15), there exists $\rho_0 \geq \max\{\rho_1, 2R\}$ such that for all $|\rho| \geq \rho_0$ and all $z \in C_k$,

$$|Q(z)| < |nz^{n-1} + \lambda|.$$

Therefore Rouché's theorem implies that, for every such ρ, the polynomial

$$f'(z) + \lambda = nz^{n-1} + \lambda + Q(z)$$

has exactly one zero inside each disk bounded by C_k. Since $f'(z) + \lambda$ has degree $n-1$, these are all of its zeros. Denote them by $c_1(\lambda), \ldots, c_{n-1}(\lambda)$. In particular,

$$c_k(\lambda) = \rho\omega_k + O(1)$$

uniformly in k as $|\lambda| \to \infty$.

Now write

$$f(z) = z^n + b_{n-1}z^{n-1} + \cdots + b_0.$$

Since $f'(c_k(\lambda)) + \lambda = 0$, we have

$$H_\lambda(c_k(\lambda)) = f(c_k(\lambda)) + \lambda c_k(\lambda) = f(c_k(\lambda)) - c_k(\lambda)f'(c_k(\lambda)).$$

But

$$f(z) - zf'(z) = -(n-1)z^n + \widetilde{Q}(z),$$

where $\deg \widetilde{Q} \leq n-1$. Hence

$$H_\lambda(c_k(\lambda)) = -(n-1)c_k(\lambda)^n + \widetilde{Q}(c_k(\lambda)).$$

Since $c_k(\lambda) = \rho\omega_k + O(1)$, it follows that

$$c_k(\lambda)^n = \rho^n\omega_k^n + O(|\rho|^{n-1}) = \rho^n\omega_k + O(|\rho|^{n-1}),$$

because $\omega_k^{n-1} = 1$ implies $\omega_k^n = \omega_k$. Also

$$\widetilde{Q}(c_k(\lambda)) = O(|\rho|^{n-1}).$$

Therefore

$$H_\lambda(c_k(\lambda)) = -(n-1)\rho^n\omega_k + O(|\rho|^{n-1}).$$

If $k \neq \ell$, then

$$H_\lambda(c_k(\lambda)) - H_\lambda(c_\ell(\lambda)) = -(n-1)\rho^n(\omega_k - \omega_\ell) + O(|\rho|^{n-1}).$$

Since $\omega_k \neq \omega_\ell$, the leading term has modulus bounded below by $c|\rho|^n$ for some constant $c > 0$ depending only on n, whereas the error term is $O(|\rho|^{n-1})$. Consequently, for all sufficiently large $|\lambda|$, the critical values $H_\lambda(c_1(\lambda)), \dots, H_\lambda(c_{n-1}(\lambda))$ are pairwise distinct.

Thus $\Delta(\lambda) \neq 0$ for all sufficiently large $|\lambda|$, so $\Delta \not\equiv 0$. Since Δ is a polynomial, the set Σ_2 is finite. Choose a small $\lambda \notin \Sigma_1 \cup \Sigma_2$. Then H_λ has simple critical points and pairwise distinct complex critical values.

Step 4: keep all critical values in K above 2α. Let

$$Z_K := \{\zeta \in K : f'(\zeta) = 0\}.$$

Since f' has no zeros on ∂K, the set Z_K is finite and contained in $\mathrm{int}(K)$. Set

$$M := \max_{z \in K} |z|.$$

If $Z_K = \varnothing$, then $f'(z) \neq 0$ for every $z \in K$. Since K is compact and $|f'|$ is continuous, the minimum

$$\delta_K := \min_{z \in K} |f'(z)|$$

is strictly positive. Choose λ so small that

$$|\lambda| < \frac{\delta_K}{2}.$$

Then for every $z \in K$,

$$|g'(z)| = |f'(z) + \lambda| \geq |f'(z)| - |\lambda| \geq \frac{\delta_K}{2} > 0.$$

Hence g' has no zeros on K. In particular, u_g has no critical points in K, so there are no critical values of u_g arising from critical points in K. Moreover, any zero of g lying in one of the root-disks D_j is a logarithmic singularity of u_g, not a smooth critical point, and by Step 1 it satisfies $u_g > 5\alpha/2$. Therefore such zeros do not contribute any level in $[\alpha, 2\alpha]$. In particular, no critical value of u_g on K lies in $[\alpha, 2\alpha]$. In this case set

$$\beta_* := 1.$$

If $Z_K \neq \varnothing$, then, because $u = -\log|f|$ has no critical values in $[0, 4\alpha)$, every $\zeta \in Z_K$ satisfies

$$|f(\zeta)| \leq e^{-4\alpha}.$$

Set

$$M_* := \max_{\zeta \in Z_K} |f(\zeta)|, \qquad \eta_* := e^{-2\alpha} - M_* > 0.$$

Choose pairwise disjoint closed disks $E_\zeta \Subset \mathrm{int}(K)$, one around each $\zeta \in Z_K$, so small that

$$|f(z)| \leq M_* + \frac{\eta_*}{4} = e^{-2\alpha} - \frac{3\eta_*}{4} \qquad (z \in E := \bigcup_{\zeta \in Z_K} E_\zeta),$$

and such that f' has no zeros on $K \setminus E$. Set

$$\delta_E := \min_{z \in K \setminus E} |f'(z)| > 0, \qquad \delta_* := \min_{\zeta \in Z_K} \min_{z \in \partial E_\zeta} |f'(z)| > 0.$$

Shrink λ further so that

$$|\lambda| < \min\left\{\frac{\delta_E}{2}, \frac{\delta_*}{2}, \frac{\eta_*}{4M}\right\}.$$

Then for every $z \in K \setminus E$,

$$|f'(z) + \lambda| \geq |f'(z)| - |\lambda| > \frac{\delta_E}{2} > 0,$$

so $f' + \lambda$ has no zeros on $K \setminus E$.

Moreover, for each $\zeta \in Z_K$ and every $z \in \partial E_\zeta$,

$$|\lambda| < |f'(z)|.$$

Hence, by Rouché's theorem on ∂E_ζ, the functions f' and $f' + \lambda$ have the same number of zeros in E_ζ, counted with multiplicity. In particular, every zero of $f' + \lambda$ lying in K belongs to E. Since the critical points of $H_\lambda(z) := f(z) + \lambda z$ are precisely the zeros of

$$H'_\lambda(z) = f'(z) + \lambda,$$

it follows that every critical point c of H_λ in K lies in E. Therefore

$$|f(c)| \leq e^{-2\alpha} - \frac{3\eta_*}{4}.$$

Hence

$$|H_\lambda(c)| \leq |f(c)| + |\lambda|\,|c| \leq |f(c)| + |\lambda| M < e^{-2\alpha} - \frac{\eta_*}{2}.$$

Now choose β so small that

$$|\beta| < \frac{\eta_*}{2}.$$

Then for every critical point c of H_λ in K,

$$|g(c)| = |H_\lambda(c) + \beta| < e^{-2\alpha}.$$

Equivalently, every critical value of u_g arising from a critical point in K is $> 2\alpha$. In particular, no critical point of u_g in K has critical value in $[\alpha, 2\alpha]$. In this case set

$$\beta_* := \frac{\eta_*}{2}.$$

Step 5: make the moduli distinct. Let $a_1, \ldots, a_N$ be the pairwise distinct complex critical values of H_λ arising from critical points in K. After adding β, they become $a_1 + \beta, \ldots, a_N + \beta$. Their moduli fail to be distinct only if, for some $i \neq j$,

$$|a_i + \beta| = |a_j + \beta|.$$

Equivalently,

$$|a_i + \beta|^2 - |a_j + \beta|^2 = 0,$$

that is,

$$2\Re\Big(\beta\overline{(a_i - a_j)}\Big) + |a_i|^2 - |a_j|^2 = 0.$$

For fixed $i \neq j$, this is a nontrivial affine real linear equation in $\beta \in \mathbb{C} \cong \mathbb{R}^2$, so it defines an affine real line. Thus the bad values of β lie in a finite union of lines. Fix the previously chosen $\lambda \notin \Sigma_1 \cup \Sigma_2$. By Step 1 there exists $\beta_1 > 0$ such that every β with $|\beta| < \beta_1$ preserves the zero configuration in the disks D_j, and by Step 4 every β with $|\beta| < \beta_*$ keeps all critical values in K above 2α. Hence, with

$$\beta_0 := \min\{\beta_1, \beta_*\} > 0,$$

we may choose β with $0 < |\beta| < \beta_0$ outside that finite union of lines. Then the critical values of u_g arising from critical points in K and larger than 2α are pairwise distinct.

This proves the lemma. $\square$

Remark. Lemma 13 is only a *local* genericity statement on the fixed compact set K. It does not claim anything about critical points of u_g outside K. This is sufficient for the argument below, because Lemmas 14 and 15 show that the relevant perturbed component V_g lies in K, so every critical point relevant to Proposition 12 is contained in K.

6. Stability of the collar integral and of the relevant component

We retain the notation of the previous section.

Lemma 14 (Component stability)**.** *Suppose $g \to f$ uniformly on K and that, for all sufficiently close g,*

(a) *each D_j contains exactly one zero of g;*
(b) *g has no zeros on $K \setminus \bigcup_j D_j$.*

Then, for all sufficiently small perturbations, there exists a unique connected component V_g of

$$\{z \in K : -\log|g(z)| > \alpha\}$$

such that

$$L_0 \subset V_g \subset \mathrm{int}(K) \subset U,$$

and V_g contains exactly the m zeros of g lying in the disks D_j.

Proof. On L_0 one has $u \geq 3\alpha/2$, while on ∂K one has $u = \alpha/2$. Uniform convergence implies

$$-\log|g| > \alpha \text{ on } L_0, \qquad -\log|g| < \alpha \text{ on } \partial K$$

for all sufficiently small perturbations. Thus the connected set L_0 lies in a unique connected component

$$V_g \subset \mathrm{int}(K)$$

of the open set

$$\{-\log|g| > \alpha\} \cap \mathrm{int}(K).$$

Since $-\log|g| < \alpha$ on ∂K, this is equivalently a connected component of $\{-\log|g| > \alpha\} \cap K$.

Fix j. Since $\partial D_j \subset L_0 \subset V_g$, we have $|g| < e^{-\alpha}$ on ∂D_j. By the maximum modulus principle,

$$|g| < e^{-\alpha} \text{ throughout } D_j.$$

Hence $D_j \subset V_g$, so the zero of g inside D_j belongs to V_g. There are no other zeros of g in K, hence V_g contains exactly these m zeros.

We next prove uniqueness. Suppose, for contradiction, that W is another connected component of

$$\{-\log|g| > \alpha\} \cap K.$$

Because $-\log|g| < \alpha$ on ∂K, every connected component of $\{-\log|g| > \alpha\} \cap K$ is contained in $\mathrm{int}(K)$; equivalently, W is a connected component of the open set

$$\{-\log|g| > \alpha\} \cap \mathrm{int}(K).$$

Since V_g contains every disk D_j, the set W is disjoint from each D_j.

We claim that in fact

$$\overline{W} \cap \partial D_j = \varnothing \qquad (1 \leq j \leq m).$$

Assume otherwise, and choose

$$p \in \overline{W} \cap \partial D_j$$

for some j. Because $\partial D_j \subset L_0 \subset V_g$, and because $L_0 \subset \mathrm{int}(K)$ while $-\log|g| > \alpha$ on L_0, there exists $r > 0$ such that

$$B(p,r) \subset \{-\log|g| > \alpha\} \cap \mathrm{int}(K).$$

Since $p \in \partial D_j$ and $D_j \subset V_g$, the ball $B(p,r)$ meets V_g. Because $p \in \overline{W}$, it also meets W. This is impossible, since a connected subset of the open set $\{-\log|g| > \alpha\} \cap \mathrm{int}(K)$ cannot meet two distinct connected components of that set. This proves the claim.

Therefore

$$\overline{W} \subset K \setminus \bigcup_{j=1}^{m} D_j.$$

By assumption, g has no zeros on $K \setminus \bigcup_j D_j$, so $-\log|g|$ is harmonic on an open neighborhood of $K \setminus \bigcup_j D_j$, hence on a neighborhood of $\overline{W}$.

We claim that

$$\partial W \subset \Big(K \cap \{-\log|g| = \alpha\}\Big) \cup \partial K.$$

Indeed, let $p \in \partial W \cap \mathrm{int}(K)$. Since W is a connected component of the open set $\{-\log|g| > \alpha\} \cap \mathrm{int}(K)$, the point p cannot satisfy $-\log|g(p)| > \alpha$; otherwise, by continuity, some ball $B(p,r) \subset \mathrm{int}(K)$ would be contained in $\{-\log|g| > \alpha\} \cap \mathrm{int}(K)$ and would meet W, forcing $B(p,r) \subset W$, contrary to $p \in \partial W$. On the other hand, p is a limit point of $W \subset \{-\log|g| > \alpha\}$, so continuity gives $-\log|g(p)| \geq \alpha$. Hence $-\log|g(p)| = \alpha$.

If instead $p \in \partial W \cap \partial K$, then the first paragraph of the proof gives $-\log|g(p)| < \alpha$. Therefore

$$-\log|g| \leq \alpha \qquad \text{on } \partial W.$$

Since W is bounded and $-\log|g|$ is harmonic on a neighborhood of $\overline{W}$, the maximum principle yields

$$-\log|g| \leq \alpha \qquad \text{throughout } W,$$

contradicting

$$W \subset \{-\log|g| > \alpha\}.$$

Thus V_g is the unique connected component of

$$\{-\log|g| > \alpha\} \cap K.$$

Finally, because $-\log|g| < \alpha$ on ∂K, a path in $\{-\log|g| > \alpha\}$ starting in V_g cannot cross ∂K. Hence V_g is also a connected component of the full superlevel set

$$\{-\log|g| > \alpha\}.$$

□

Lemma 15 (Continuity of the collar integral)**.** *Assume $g \to f$ in $C^1(K)$, and assume that, for all sufficiently close g,*

(a) *each D_j contains exactly one zero of g;*
(b) *g has no zeros on $K \setminus \bigcup_{j=1}^{m} D_j$.*

Let V_g be the corresponding component from Lemma 14, and define

$$q_g := \frac{1}{2\pi}\int_{\alpha}^{2\alpha} P_g(t)\,dt,$$

where

$$P_g(t) := \mathcal{H}^1(\{z \in V_g : u_g(z) = t\}) \qquad \text{for regular } t,$$

and $u_g := -\log|g|$. Then

$$q_g \to q \qquad \text{as } g \to f \text{ in } C^1(K).$$

In particular, for all sufficiently small perturbations,

$$q_g > \frac{q}{2}.$$

Proof. Because the disks D_j were chosen inside $\{u > 5\alpha/2\}$, the compact band

$$B := \{z \in U : \alpha/2 \le u(z) \le 5\alpha/2\}$$

is disjoint from $\bigcup_{j=1}^m D_j$. Since all zeros of g in K lie in the disks D_j, the function $u_g = -\log|g|$ is well defined on B for all sufficiently small perturbations.

Because f has no zeros on

$$C := K \setminus \bigcup_{j=1}^m \operatorname{int}(D_j),$$

there exists $m_C > 0$ such that

$$|f| \ge m_C \qquad \text{on } C.$$

Since $D_j \subset \{u > 5\alpha/2\}$ for each j, we also have

$$M_D := \max_{z \in \bigcup_{j=1}^m D_j} |f(z)| \le e^{-5\alpha/2} < e^{-2\alpha}.$$

Choose the perturbation so small that

$$\|g - f\|_{L^\infty(K)} < \min\{m_C/2,\, e^{-2\alpha} - M_D\}.$$

Then g has no zeros on C, and in fact

$$|g| \ge m_C/2 \qquad \text{on } C,$$

while on the union of the root disks we have

$$|g| \le M_D + \|g - f\|_{L^\infty(K)} < e^{-2\alpha}.$$

Equivalently,

$$u_g > 2\alpha \qquad \text{on } \bigcup_{j=1}^m D_j.$$

Hence the entire collar region $\{\alpha < u_g < 2\alpha\} \cap K$ is disjoint from the disks D_j.

On the zero-free compact set C, the formulas

$$u = -\log|f|, \qquad u_g = -\log|g|,$$

$$\nabla u = -\nabla(\log|f|), \qquad \nabla u_g = -\nabla(\log|g|)$$

and the lower bound $|g| \ge m_C/2$ show that $g \to f$ in $C^1(K)$ implies

$$u_g \to u \quad \text{uniformly on } C, \qquad \nabla u_g \to \nabla u \quad \text{uniformly on } C.$$

In particular, since $B \subset C$, we have uniform convergence on B as well. Since u has no critical points on B, there exists $c_0 > 0$ such that

$$|\nabla u| \geq c_0 \qquad \text{on } B.$$

Hence, for all sufficiently small perturbations,

$$|\nabla u_g| \geq c_0/2 \qquad \text{on } B,$$

so u_g has no critical points on B.

We next identify the relevant collar region for u_g. By Lemma 14, $V_g \subset K$ and V_g is the unique connected component of $\{u_g > \alpha\} \cap K$. Since $u_g \to u$ uniformly on C, for all sufficiently small perturbations we have

$$\|u_g - u\|_{L^\infty(C)} < \alpha/2.$$

Therefore, if $z \in K$ satisfies $\alpha < u_g(z) < 2\alpha$, then $z \notin \bigcup_j D_j$, so $z \in C$, and hence

$$\alpha/2 < u(z) < 5\alpha/2.$$

Therefore $z \in B$. Also, any such z belongs to $\{u_g > \alpha\} \cap K$, hence to the unique component V_g. Thus

$$\{z \in V_g : \alpha < u_g(z) < 2\alpha\} = \{z \in K : \alpha < u_g(z) < 2\alpha\} \subset B.$$

Likewise,

$$\{z \in U : \alpha < u(z) < 2\alpha\} = \{z \in K : \alpha < u(z) < 2\alpha\} \subset B.$$

Applying the coarea formula on the band B, we obtain

$$2\pi q = \int_B 1_{\{\alpha < u < 2\alpha\}} \, |\nabla u| \, dA,$$

and

$$2\pi q_g = \int_B 1_{\{\alpha < u_g < 2\alpha\}} \, |\nabla u_g| \, dA.$$

Set

$$E := \{z \in B : \alpha < u(z) < 2\alpha\}, \qquad E_g := \{z \in B : \alpha < u_g(z) < 2\alpha\}.$$

Because $u_g \to u$ uniformly on B, we have

$$1_{E_g}(z) \to 1_E(z)$$

for every $z \in B$ such that $u(z) \notin \{\alpha, 2\alpha\}$. The exceptional set is contained in

$$\{u = \alpha\} \cup \{u = 2\alpha\},$$

which has planar measure zero by Lemma 3. Thus $1_{E_g} \to 1_E$ almost everywhere on B.

Also, for g sufficiently close to f, uniform convergence of the gradients gives a constant $M_B < \infty$ such that

$$|\nabla u_g| \leq M_B \qquad \text{on } B.$$

Therefore the integrands

$$1_{E_g} |\nabla u_g|$$

are dominated by the integrable function $M_B 1_B$, and they converge almost everywhere on B to

$$1_E |\nabla u|.$$

By dominated convergence,

$$q_g \to q.$$

In particular, $q_g > q/2$ for all sufficiently small perturbations. □

7. Proof of the main theorem

We now complete the argument.

Lemma 16 (A tree lemma)**.** *Let T be a finite tree of total length L, and let $x_1, \dots, x_M$ be marked points of T. Then some pair x_i, x_j is joined by a subpath of length at most $2L/M$.*

Proof. Subdivide edges at the marked points; this does not change the total length and turns the marked points into marked vertices. Take the minimal subtree spanning the marked vertices. Every leaf of this subtree is marked. Doubling every edge produces an Eulerian graph of total length $2L$. Traversing an Euler tour and recording the first appearances of the marked vertices gives a cyclic ordering of the M marked vertices. The M gaps between consecutive marked vertices along the tour have total length $2L$, so one of them has length at most $2L/M$. The corresponding unique path in the original tree has at most that length. □

Proof of Theorem 1. If $n = 1$, then the unique root may be joined to itself by the constant curve, so the conclusion is immediate. If f has a multiple zero, the conclusion is also immediate. Thus we may assume that f is squarefree and $n \geq 2$.

By the component lemma, there exists a connected component $U \subset \Lambda_f$ containing exactly $m \geq 2$ zeros of f. Let $u = -\log|f|$ on U, let α and q be as in Section 3, and choose the root disks $D_1, \dots, D_m$ as in Section 5, so small that

$$\sum_{j=1}^{m} \operatorname{diam}(D_j) < \frac{q}{4m}. \tag{17}$$

Choose (λ, β) as in Lemma 13 so small that

$$\|g_{\lambda,\beta} - f\|_{C^1(K)} < \varepsilon_0,$$

where $\varepsilon_0 > 0$ is small enough for Lemmas 14 and 15 to apply, and set

$$g := g_{\lambda,\beta}.$$

By Lemma 13(1), each disk D_j contains exactly one simple zero b_j of g, and g has no zeros on

$$K \setminus \bigcup_{j=1}^{m} D_j.$$

Because the disks D_j lie in $\{u > 5\alpha/2\}$, they are disjoint from the fixed band

$$B = \{z \in U : \alpha/2 \leq u \leq 5\alpha/2\},$$

so g has no zeros on B as well. Let V_g be the corresponding component from Lemma 14. Then V_g is a connected component of the full superlevel set

$$\{-\log|g| > \alpha\},$$

it satisfies $V_g \subset \operatorname{int}(K) \subset U$, it contains exactly the m perturbed roots $b_1, \dots, b_m$ lying in the disks D_j, and by Lemma 15 we have

$$q_g > q/2.$$

Applying Corollary 5 with

$$h = g, \qquad v = u_g := -\log|g|, \qquad c = \alpha, \qquad W = V_g,$$

gives

$$\frac{1}{2\pi}\int_{\alpha}^{\infty} P_g(t)\,dt \le m.$$

Hence

$$\frac{1}{2\pi}\int_{2\alpha}^{\infty} P_g(t)\,dt = \frac{1}{2\pi}\int_{\alpha}^{\infty} P_g(t)\,dt - \frac{1}{2\pi}\int_{\alpha}^{2\alpha} P_g(t)\,dt \le m - q_g < m - \frac{q}{2}.$$

Since all zeros of g in V_g are the simple roots $b_1, \dots, b_m$, Proposition 12 applies to $u_g = -\log|g|$ on V_g with base level $c = \alpha$. By Lemma 14 we have $V_g \subset K$, so every critical point relevant to Proposition 12 lies in K. Therefore Lemma 13(2) gives the absence of critical values in $[\alpha, 2\alpha]$, while Lemma 13(3)–(4) give the Morse and distinct-critical-value hypotheses above 2α. Choose

$$\varepsilon := \frac{q}{8}.$$

Then Proposition 12 yields a finite tree $G \subset V_g \subset U$ containing the m perturbed roots and satisfying

$$\operatorname{len}(G) \le \frac{1}{2\pi}\int_{2\alpha}^{\infty} P_g(t)\,dt + \frac{q}{8} < m - \frac{q}{2} + \frac{q}{8} = m - \frac{3q}{8}.$$

By Lemma 16, some pair of perturbed roots, say $b_r \in D_r$ and $b_s \in D_s$, is joined in G by a subpath of length at most

$$\frac{2}{m}\operatorname{len}(G) < \frac{2}{m}\left(m - \frac{3q}{8}\right) = 2 - \frac{3q}{4m}. \tag{18}$$

The indices $r \ne s$, because the m perturbed roots are distinct and each disk D_j contains exactly one of them.

Connect b_r to the original zero $a_r \in D_r$ and b_s to the original zero $a_s \in D_s$ by straight line segments. Because each D_j is an ordinary Euclidean disk contained in U, these straight segments lie in

$$D_r \cup D_s \subset U \subset \Lambda_f.$$

The total additional length is at most

$$\operatorname{diam}(D_r) + \operatorname{diam}(D_s) \le \sum_{j=1}^{m} \operatorname{diam}(D_j) < \frac{q}{4m}$$

by (17). Therefore the resulting curve in $U \subset \Lambda_f$ joining two zeros of f has total length strictly less than

$$\left(2 - \frac{3q}{4m}\right) + \frac{q}{4m} = 2 - \frac{q}{2m} < 2.$$

This proves the theorem. □

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